



Compensation disclosures and strategic commitment: Evidence from revenue-based pay[☆]

Matthew J. Bloomfield

The Wharton School of the University of Pennsylvania, 1325 Steinberg-Dietrich Hall, 3620 Locust Walk, Philadelphia, PA 19104-6365, USA

ARTICLE INFO

Article history:

Received 30 May 2018

Revised 5 June 2020

Accepted 5 July 2020

Available online 10 April 2021

JEL classification:

D22

D80

J33

L13

M48

Keywords:

Strategic delegation

Oligopoly

Commitment

Compensation

Disclosure

ABSTRACT

A 2006 rule change in the United States mandated that publicly traded firms provide more detailed disclosures about executives' compensation plans. In response to the new disclosure requirements, Cournot firms with large market shares add revenue-based pay to their CEOs' pay packages. This change in pay practices coincides with a shift towards more aggressive product market equilibria, characterized by greater production expenditures and lower margins. Jointly, these patterns are consistent with predictions from the theory of "strategic delegation," and suggest that the new disclosure requirements enhanced the viability of committing through executive incentives. After adopting the new disclosure requirements, many firms appear to restructure their executives' pay packages as strategic devices designed to make rivals curtail their competitive actions.

© 2021 Published by Elsevier B.V.

1. Introduction

Analytical work on "strategic delegation" shows that Cournot competitors can improve their profitability by using revenue-based pay to commit to more aggressive product market behavior, thereby reducing competition from rivals (e.g., [Fershtman and Judd, 1987](#)). This tactic (hereafter the "Fershtman and Judd strategy") is more effective when agents' incentives are more clearly and credibly relayed to rivals ([Katz, 1991](#)). I exploit the introduction of the Compensation Discussion and Analysis ("CD&A") section of the proxy statement, which required public firms on American stock exchanges to provide more detailed disclosures regarding executives' compensation plans, and provide novel empirical evidence regarding public companies' use of the Fershtman and Judd strategy.

I document that many firms add revenue-based pay to their CEOs' pay packages immediately after adopting the

[☆] I am grateful for all the thoughtful guidance from Bill Schwert (the editor), an anonymous referee, my dissertation committee: Christian Leuz (my chair), Sanjog Misra, Haresh Sapra, and Doug Skinner. I would also like to thank Mark Bagnoli, Simcha Barkai, John Barrios, Phil Berger, Mary Billings, Robert Bloomfield, Matthias Breuer, Ben Charoenwong, Hans Christensen, Carlos Corona, Øystein Dajord, Jean-Pierre Dubé, Robbi Fox, Pingyang Gao, Joseph Gerakos, Brandon Gipper, Wayne Guay, Steven Kaplan, Anya Kleymenova, Steven Levitt, Mike Minnis, Valeri Nikolaev, Canice Prendergast, Robert Sanders, Amit Seru, Andrei Shleifer, Chad Syverson, Abbie Smith, Sorabh Tomar, Rodrigo Verdi, Anastasia Zakolyukina, and Frank Zhou for their many helpful insights. Lastly, I would like to thank workshop participants at the University of Chicago, Indiana University, the University of North Carolina, the Massachusetts Institute of Technology, the University of Michigan, the University of Pennsylvania, Stanford University, Harvard University, the University of California at Los Angeles, the 2016 Carnegie Mellon Accounting Mini-Conference, and the 2017 Yale Fall Accounting Conference for their feedback.

E-mail address: mjbloom@wharton.upenn.edu

new CD&A requirements. The sudden addition of revenue-based pay is most prevalent among firms that stand to benefit most from a credible commitment to aggressive product market behavior. I also find evidence to suggest that this tactic affects product market outcomes. The shift towards revenue-based pay coincides with a shift towards more aggressive product market equilibria, characterized by higher production expenditures and lower margins. Collectively, this evidence is consistent with the Fershtman and Judd strategy and suggests that strategic considerations shape executive pay; after the introduction of the CD&A, many CEOs' incentive plans appear to have been restructured as strategic devices designed to curtail competition from rivals.

Fershtman and Judd (1987) provide the model that is most closely linked to my study. They analyze a two-period Nash game in which principal–agent pairs compete to sell a homogeneous good. In the first period, profit-seeking principals simultaneously choose the relative weight on profits versus revenues in their agents' compensation contracts. In the second period, agents take all contracts as given and compete on quantity to maximize their own contractually defined payoffs. The model implies that principals will rationally place a positive weight on revenue, but this weight diminishes as the number of product market competitors grows. Intuitively, revenue-based pay incentivizes an agent to overproduce relative to the profit-maximizing quantity. This commitment to aggressive behavior disciplines rival agents, causing them to curtail their own production and thereby exert less downward pressure on prices.

The viability of this strategy depends on a principal's ability to clearly and credibly reveal her agent's incentives; the better the principal can relay her agent's incentives, the greater the benefits of using the strategy (Katz, 1991). Jointly, Fershtman and Judd (1987) and Katz (1991) provide the following testable prediction: more clear and credible disclosure of an agent's incentives will lead to greater use of revenue-based pay. However, this prediction depends crucially on two factors: competitive mode and market share.

Regarding the competitive mode, the above prediction requires that rival firms' strategic actions be substitutes, as opposed to complements. Strategic actions are substitutes if one firm's more aggressive behavior begets less aggressive behavior from its rivals (i.e., best response functions are negatively sloped).¹ If strategic actions are complements (i.e., best response functions are positively sloped), firms will not endeavor to commit to more aggressive behavior as doing so would invite more aggressive behavior from rivals. Hereafter, I refer to industries as “Cournot” if strategic actions are substitutes and “Bertrand” if strategic actions are complements.² My primary measure for

the competitive mode is constructed using the approach developed by Kedia (2006). This method of classification is based on the relation between a firm's marginal returns from aggressiveness and a representative rival firm's aggressiveness—if the relation is negative (positive), then strategic actions are said to be substitutes (complements). As there is no widely accepted method of classifying industries as Cournot versus Bertrand, I supplement the Kedia (2006) measure with seven alternatives and find that they yield similar inferences.³

Regarding market share, the above prediction requires that a firm command enough of the product market that its actions materially influence rivals' actions, but not so much that there is no competition worth curtailing. In an oligopoly with only a few large players, each firm wields considerable influence and commitment confers substantial benefits. In a more competitive market where each firm is approximately an atomistic price taker, no single firm can elicit a significant response from its rivals and commitment yields no major benefits. In an industry with both large and small players, the large players benefit most from commitment. In the extreme case of a monopolist, the benefits of strategic commitment vanish as there are no rivalrous actions to curtail.⁴ I use a firm's pre-CD&A market share as a firm-level measure of the extent to which the firm can benefit from strategic commitment. This measure captures variation both across industries (e.g., number of firms in the market) and within industries (e.g., relative size of a firm).

In my empirical tests, I exploit the mandated introduction of the CD&A as a plausibly exogenous shock to pay package disclosure quality and examine how firms react to the new requirements as a function of market share and competitive mode. The key supposition underlying my research design is that the CD&A led to an improvement in firms' ability to clearly and credibly convey managerial incentives and thereby enhanced the viability of the Fershtman and Judd strategy. My design does not require that firms were entirely unable to use executives' incentive contracts as commitment devices prior to the CD&A, nor that firms were perfectly able to do so after the CD&A. My design only requires that, on the margin, the CD&A led to an improvement in the viability of committing through executive pay.

In Cournot industries, I find that the prevalence of revenue-based pay in CEO incentive plans increases suddenly after CD&A adoption.⁵ This shift in revenue-based pay is driven by more oligopolistic firms (i.e., those with large—but not overwhelmingly large—market shares). Of Cournot firms with market shares between 10% and 65%, reliance on revenue-based pay more than doubles (increas-

¹ For example, if firms compete by choosing production quantities (Cournot, 1838), then one firm's choice to produce more pushes rivals to produce less as a best response; their actions are strategic substitutes.

² Firms need not compete on quantities (prices) in order for their strategic actions to be substitutes (complements). A wide variety of games beyond Cournot and Bertrand can produce similar interdependencies. I use the terms “Cournot” and “Bertrand” loosely, referring to any product

market in which rivalrous firms' strategic actions are substitutes versus complements.

³ A detailed description of all eight measures, and the results they produce, can be found in the Appendix.

⁴ For the purposes of this study, I do not consider competitive threats of entry. An incumbent monopolist may indeed benefit from strategic commitment to maintain monopolistic rents by deterring entry.

⁵ Throughout this study, I focus exclusively on CEO pay. Going forward, all mentions of “pay” refer specifically to CEO pay.

ing by roughly 20 percentage points) in the first year of the post-CD&A period. More atomistic or monopolistic firms do not appear to shift towards revenue-based pay. In Bertrand industries, I find no sudden changes in response to the CD&A at all. Bertrand firms trend towards greater reliance on revenue-based pay over my sample period, but the more oligopolistic players are less likely than their atomistic or monopolistic peers to add revenue-based pay after the CD&A.

Collectively, this descriptive evidence is consistent with my theoretical predictions. The primary empirical challenge lies in establishing whether strategic considerations drive the patterns in revenue-based pay that I document. I rely on an array of empirical tactics to assess alternative explanations for these patterns and find that the evidence supports a “strategic commitment” interpretation. I outline several of these tactics below.

First, in all of the regression analyses, I use firm and year fixed effects to identify coefficients from within-firm and within-year variation in the use of revenue-based pay. This structure allows each firm to act as its own control and ensures that my results are immune to bias arising from time invariant cross-sectional heterogeneity, evolving sample composition or broad time trends.

Second, I augment the baseline regression analyses with several refinements to better establish the strategic impetus underlying the observed pay practice changes: (1) I control for firm size and find that market share explains Cournot firms’ shifts towards revenue-based pay, while firm size does not; (2) I exploit the staggered roll-out of the CD&A and find that pay practice changes occur tightly around firms’ own CD&A adoption dates—the increased usage of revenue-based pay begins in fiscal 2006 for 2006 adopters (December year-end firms) and in fiscal 2007 for 2007 adopters (September through November year-end firms); and (3) I use coarsened exact matching to ensure that the Cournot and Bertrand firms in my sample are approximately identical with respect to known agency-theoretic determinants of revenue-based pay—the changes in pay practices become even more stark after matching.

Third, I examine firms’ usage of other supplemental performance metrics to assess whether the observed patterns in revenue-based pay are actually just a microcosm of a broad overhaul in executive pay practices. I find no evidence that large-share Cournot firms differentially changed their use of other types of performance metrics; the observed patterns appear to be specific to revenue-based pay. Moreover, I control for firms’ reliance on other types of performance metrics and find that doing so does not affect my inferences.

Fourth, I leverage the predicted non-monotonic effect of market share on strategic behavior. A firm benefits most from strategic commitment when it commands a substantial portion of the market but still faces a great deal of competition (i.e., oligopolistic competitors). Once a firm garners an overwhelming share of the market (i.e., a monopolist), strategic commitment loses much of its value. Consistent with a strategic commitment interpretation, I find clear evidence of the predicted non-monotonicity. With respect to explaining Cournot firms’ shift towards revenue-based pay, the marginal effect of market share

switches from significantly positive to significantly negative in the right tail of the market share distribution. This non-monotonicity is difficult to explain without an appeal to strategic considerations.

Lastly, I examine product market outcomes to assess whether they change after the CD&A in the predicted manner. Consistent with [Fershtman and Judd \(1987\)](#), I find that post-CD&A product market outcomes shift towards more aggressive equilibria in highly concentrated Cournot industries. Coincident with the shift towards revenue-based pay, firms appear to ramp up production and drive down prices.

One might be concerned that the documented shifts in revenue-based pay do not reflect actual changes in executive incentives but merely changes in *disclosed* executive incentives. That is, perhaps firms’ reliance on revenue-based pay is largely constant, but firms would not reliably disclose its use until after the CD&A. While I cannot definitively rule this out, I take several steps to address this possibility.

First, I exclude from my sample any firm-year observations for which Incentive Lab (the primary data source used in my analysis) lists no performance metrics or lists at least one metric as “Vague.” Second, I calculate the implied pre-CD&A non-disclosure rate, under the assumption that the pay packages remained unchanged in the first year of the post-CD&A period. I assume that all apparent changes in compensation practices were driven by the disclosure of previously undisclosed performance metrics, not actual changes in compensation practices. Under this assumption, I find that the magnitude of the disclosure rate issue would have to be both very severe and highly specific to large-share Cournot firms’ use of revenue-based pay. Third, I examine the rate at which firms remove revenue-based pay from their CEOs’ pay packages, and find that removal rates also increase around CD&A adoption. This pattern cannot be explained by pre-CD&A non-disclosure.

This study contributes to multiple streams of literature. First and foremost, this paper contributes to the empirical literature on strategic delegation as the first paper to provide evidence that firms use their executive pay packages as strategic commitment devices.⁶ I utilize detailed grant-level compensation data and a quasi-experimental design to identify the use of incentive contracts as commitment devices. I find that, immediately after the introduction of a pay package disclosure mandate, firms are more likely to add revenue-based pay to their CEOs’ pay packages if the benefits of committing to aggressive product market behavior are greater.

Second, this study contributes to the emerging literature on the role of public disclosure in shaping prod-

⁶ [Aggarwal and Samwick \(1999\)](#) and [Kedia \(2006\)](#) provide some of the only related empirical work, both providing suggestive evidence that oligopolistic interdependencies shape executive pay. [Kedia \(2006\)](#) shows that pay for performance sensitivity is higher in Bertrand industries, while [Aggarwal and Samwick \(1999\)](#) find that in more competitive industries, managers’ bonuses are more highly correlated with rival firm performance. However, both results are based purely on cross-sectional correlations, and neither paper exploits any sort of plausibly exogenous variation in the contracting environment (e.g., disclosure regulation). Thus, their descriptive results, while intriguing, cannot be confidently interpreted as evidence of strategic delegation.

uct market competition. This is one of several recent papers to provide empirical evidence suggesting that firms use public disclosure strategically, as a mechanism to curtail competitive pressures from current and potential rivals. For example, Bertomeu et al. (2021), Kepler (2021), and Bourveau et al. (2020) provide evidence that firms use voluntary disclosure to facilitate tacit collusion among rivals. Burks et al. (2018), Bloomfield and Tuijn (2019), and Glaeser and Landsman (2019) show that firms use voluntary disclosure to deter product market entry. This study complements existing work by providing evidence that firms use executive pay disclosures to curb competition from incumbent rivals.

Third, this study contributes to the literature on the link between executive compensation and firm behavior. Prior work has focused predominantly on the impact of equity incentives on risk-taking behavior (e.g., Knopf et al., 2002; Coles et al., 2006; Low, 2009; Dittmann et al., 2017; Shue and Townsend, 2017; Baldenius and Michaeli, 2017; Baldenius and Michaeli, 2019). This paper is the first to provide broad sample empirical evidence suggesting that firms behave more aggressively when the CEO is incentivized with revenue-based pay.⁷

Fourth, this study contributes to the budding literature on cost shielding in executive pay plans. In a contemporaneous study, Bloomfield et al. (2020) analyze firms' use of cost shielding as a tool for incentive alignment, showing that revenue-based pay can help align the incentives between shareholders and executives by mitigating managerial myopia. They find that this tactic is particularly useful in younger firms, as well as firms for which intangible investments (i.e., R&D and advertising) are more important. Our two studies complement each other, showing both strategic and non-strategic roles for revenue-based pay in executive incentive plans.

Lastly, it is worth relating my study to the growing literature on common ownership. In particular, Antón et al. (2020) provide evidence that common ownership affects managerial incentives. On the surface, our two studies are similar. Both show that product market considerations—specifically imperfect competition—can affect the nature of executive pay by creating justification for incentives other than profit maximization. However, Antón et al. (2020) motivate their analysis from agency theory. They examine a setting in which principals own shares of natural rivals and show that executive compensation can be adjusted to better align the incentives of managers and principals. In contrast, I examine the role of executive compensation in facilitating strategic commitment, not incentive alignment.⁸ Moreover, our two studies do not merely provide two different theories for the same empirical patterns; we predict and find distinct empirical results.

My study focuses on revenue-based pay and the role of pay disclosures, while Antón et al. (2020) focus on pay-for-performance sensitivities and the role of common ownership.

The remainder of this paper is organized as follows. In Section (2), I develop my empirical predictions; in Section (3), I provide a summary of the institutional setting used to test my predictions; in Section (4), I outline my data sources and describe the sample selection and variable construction procedures; in Section (5), I present the findings; and in Section (6), I conclude. In the Appendix, I describe my alternative measures for competitive mode, and in the Internet Appendix, I present a model of strategic delegation and additional tabulated analyses.

2. Hypothesis development and discussion

In this section, I discuss related theoretical literature, and develop my testable predictions. I also discuss the connection to traditional contracting theory, à la Holmström (1979).

2.1. Testable predictions

In many oligopolistic industries, firms stand to gain by credibly committing to aggressive product market behavior, thereby deterring entry or discouraging incumbent rivals' competitive actions. Aggressive commitment can take many forms. Firms can build up large capacities (e.g., Kreps and Scheinkman, 1983; Moreno and Ubeda, 2006), invest in marginal cost reductions (e.g., Brander and Spencer, 1983; Dixit, 1980; Raith, 2003), engage in “divisionalization” (e.g., Schwartz and Thompson, 1986; Veendorp, 1991), integrate with customers or suppliers (e.g., McGuire and Staelin, 1983; Hart et al., 1990) or “leverage up” (e.g., Brander and Lewis, 1986, 1988). Fershtman (1985) and Vickers (1985) introduce an alternative way for a firm to commit: “strategic delegation.”

Fershtman and Judd (1987) extend the works of Fershtman (1985) and Vickers (1985) by analyzing the strategic role of revenue-based pay in an n -firm Cournot oligopoly. They show that firms in Cournot industries can increase their profitability by using revenue-based pay to commit to more aggressive product market behavior. The more of the market a firm commands, the greater the benefits of doing so. However, the viability of this strategy depends crucially on how clearly and credibly the principal can reveal her agent's incentives to rivals (Katz, 1991). Hence, I offer the following:

Prediction 1: In Cournot industries, firms with higher market shares will differentially shift towards revenue-based pay when required to credibly disclose their CEOs' pay packages.

Thus far, I posit a positive relation between market share and the incentives to behave strategically. For the vast majority of the firms in my sample, this is likely to be the case. Intuitively, a firm with a small market share is approximately atomistic and does not stand to gain from strategic commitment because its actions will not affect other firms' behavior. In contrast, a major product market

⁷ In a contemporaneous working paper, Bloomfield et al. (2021) provide related empirical evidence showing that firms behave more aggressively when the CEO is compensated on the basis of relative performance.

⁸ Strategic delegation works by intentionally misaligning an agent's incentives from those of the principal. It is the divergence, rather than congruence, of their incentives that provides strategic gains. By aligning the agent's incentives to her own, a principal would reduce agency costs, but also forgo the potential strategic benefits of delegation.

player with a considerable market share can benefit significantly from strategic play because rational rivals will alter their own behavior as a best response.

While this logic holds in most situations, it does not hold in the limit. The incentives to behave strategically depend jointly on: (1) having sufficient market share to be able to influence rivals; and (2) facing enough competition to care about curtailing it. In the right tail of the market share distribution, a firm may stop behaving strategically as it no longer faces enough competition. With respect to strategic commitment, nearly monopolistic firms are likely to behave less like powerful oligopolists and more like atomistic small-share firms.⁹ Hence, I offer the following:

Prediction 2: Among Cournot competitors, the post-disclosure shift toward revenue-based pay will be non-monotonic in market share.

I do not formulate an ex ante prediction regarding the exact point at which the marginal effect of market share changes sign. As a rough benchmark, I look to the symmetric case analyzed by Fershtman and Judd (1987). They find that the maximum weight on revenue occurs in duopolies and triopolies (i.e., market shares of 33.3% to 50%). With a slight abuse of their model, one can calculate that the weight on revenue-based pay is maximized at a market share of $\frac{1}{1+\sqrt{2}} \approx 41.4\%$, which corresponds roughly to the 90th percentile in my sample.¹⁰

Predictions 1 and 2 relate to whether or not firms will alter their executives' pay packages. Such changes would likely affect product market outcomes. Accordingly, I offer the following:

Prediction 3: In concentrated Cournot product markets, the shift towards revenue-based pay will manifest in more aggressive product market equilibria, with higher levels of production and lower margins/markups.

My predictions focus on Cournot firms. It is not ex ante clear that the opposite predictions should apply to Bertrand firms; the theoretical predictions vis-à-vis revenue-based pay are not as clear. If Bertrand firms were to use executive incentives to facilitate strategic commitment, they would likely be better served by using margin-based bonuses or cost-based penalties. In my empirical analyses, I consider Cournot firms to be the focal sample and use Bertrand firms only as a benchmark for

comparison—if the same patterns were to arise for both modes of competition, it would suggest that strategic considerations are not responsible.

2.2. Discussion

The central tension in traditional contracting theory (i.e., “agency theory”) is the misalignment of incentives between principal and agent. Incentive misalignment can take many forms: private costs or benefits, differential risk preferences, divergent time horizons, etc. Under this paradigm, the separation of ownership and control is a problem that must be solved, and an agency cost minimizing contract is the optimal, but typically imperfect, solution.¹¹

I build upon Schelling's (1960) theoretical framework of strategic delegation to explore the complementary perspective that owners can exploit the separation of ownership and control for strategic gain. Under this paradigm, divergent incentives between principal and agent are not a problem to be solved, but a source of strategic value. When multiple rivalrous agents' actions are interdependent, one agent's incentives affect not only that agent's actions but also those of rivals. If an agent's incentives can be observably altered by a compensation contract, such alterations can be used strategically to manipulate rivals' behavior (e.g., Fershtman, 1985; Vickers, 1985; Sklivas, 1987; Fershtman and Judd, 1987).¹²

Agency theory and strategic delegation are not mutually exclusive. Pay packages can be designed to facilitate both incentive alignment and strategic commitment. The purpose of this study is not to horse-race agency theory and strategic delegation against each other. Rather, the goal is to hold fixed the agency-theoretic drivers of compensation and examine whether and how firms adjust their contracts when the viability of strategic contracting is exogenously shocked by a pay package disclosure mandate.

3. Setting: the introduction of the CD&A

In 2006, the US Securities and Exchange Commission (SEC) passed a new rule requiring publicly listed firms to provide much more detailed summaries and discussion of their compensation practices for top executives. These new disclosures form the CD&A section of a firm's annual proxy statement. In addition to disclosing a list of performance metrics, and the total amount of compensation awarded to each executive (as had been required for decades), firms

⁹ This result manifests clearly in Fershtman and Judd (1987). They show that the equilibrium weight on revenue is zero in the case of a monopolist (i.e., $n = 1$) and also in the case of perfect competition (i.e., as $n \rightarrow \infty$). The result is also present in the more flexible model, presented in Section IA1 of the Internet Appendix. The value of strategic commitment is determined jointly by influence, $\frac{\partial q_i}{\partial q_j}$ (i.e., how much firm i 's action affects the representative rival) and by competitive pressure, $\frac{\partial \pi_i}{\partial q_j}$ (i.e., how much firm i cares about the representative rival).

¹⁰ In their equation (17), Fershtman and Judd (1987) characterize the equilibrium weight on revenue as $\frac{(n-1)(a\mu-\mu^2-\sigma^2)}{(n^2+1)(\mu^2+\sigma^2+1)}$, where n is the number of firms in the market, a is the level of demand, μ is the expected marginal cost, and σ is the standard deviation of marginal cost. Taking a first order condition reveals that, irrespective of a , μ and σ , the equilibrium weight on revenue is maximized for $n = 1 + \sqrt{2}$, which corresponds to a market share of $\frac{1}{1+\sqrt{2}} \approx 41.4\%$. I call this an “abuse” of the model because the equation was derived by assuming a symmetric market with n firms; this assumption cannot apply to a market with a non-integer number of firms.

¹¹ See, for example: Jensen and Meckling (1976); Holmström (1979); Baiman and Demski (1980); Holmström and Milgrom (1987, 1991, 1994); Lambert and Larcker (1987).

¹² Koçkesen and Ok (2004) and Koçkesen (2007) depart from this literature by endogenizing the choice to delegate decision rights. In their framework, delegation can be used strategically, even if the contracts are not disclosed, because the decision to delegate functions as a costly signal (à la Spence (1973)), which reveals the contract. In the context of my study, it seems unlikely that the owners of the largest publicly traded firms base their decisions whether or not to delegate decision rights to executives on strategic incentives. Accordingly, I assume that the decision to delegate is an exogenous precondition and analyze the characteristics, rather than the existence, of compensation contracts.

were further required to disclose how they determined the amount of compensation. How much weight was placed on each metric? What were the performance targets for each metric, and how large were the payouts for achieving said goals? That is, the CD&A provides a mandated public disclosure about the explicit monetary incentives provided to a firm's top management team. Consistent with these requirements, [Gipper \(2021\)](#) finds that the CD&A provides “a narrative description of the policies and determinants of compensation, with much greater details about the incentive structure” ([Gipper, 2021](#), page 2). Publicly listed firms are required to comply with the new rules starting with the first fiscal year ending after December 15, 2006. I exploit the introduction of the CD&A as a plausibly exogenous shock to the viability of committing through executive pay.

One might legitimately question why firms would wait until the CD&A to use the Fershtman and Judd strategy. Given the benefits of strategic commitment, perhaps firms would use this strategy prior to the CD&A by disclosing the details of their compensation contracts voluntarily. First and foremost, I do not assume that all firms waited until the CD&A to use this strategy. It is quite possible that some firms were able to use voluntary disclosures to implement the Fershtman and Judd strategy in the pre-CD&A period. Without strong assumptions about the characteristics of the optimal non-strategic incentive plan, I cannot identify whether firms were using strategic contracts prior to the CD&A. My predictions are based on the notion that the CD&A made the strategy more effective, thereby promoting its use.

There are several mechanisms through which the CD&A could improve the viability of the Fershtman and Judd strategy. One possibility is that pay package disclosures were simply not salient enough in the pre-CD&A period. If it was understood that firms would not bother to look closely at rivals' Compensation Committee Reports, then strategically altering a CEO's pay package would carry all of the downsides of a misaligned manager without any of the strategic benefit. With the introduction of the CD&A came more attention, both from investors and product market rivals, which could improve the net benefits of strategic incentive contracting.

A related possibility is that voluntarily disclosed contract details were not sufficiently credible to elicit a response from rivals. Consistent with this explanation, [Morse et al. \(2011\)](#) study compensation paid to CEOs over the 1993–2003 period (before the CD&A's introduction), and find evidence that firms routinely deviate from disclosed pay plans. They argue that such deviations are the result of rent-seeking behavior by powerful managers, and that more credible disclosure was needed to prevent these ex post deviations. On the other side of the coin, [Gipper \(2021\)](#) documents that executive pay levels rise following the CD&A and argues that this might be because the CD&A disclosure can “bind” the compensation committee to a pay plan, and that binding to a set of performance metrics ex ante might become inefficient when the committed compensation determinants do not best measure the efforts of management, ex post.

While the two studies differ in their normative implications, they are aligned in suggesting that mandatory disclosures, as required under the CD&A, are more effective than voluntary disclosures at binding firms to a pay plan and credibly conveying executive incentives. This view is shared by industry experts with Exequity's Robbi Fox commenting:

“Historically, compensation committees have deviated from their stated goals to provide executives with discretionary bonuses, but it's become increasingly rare. That type of behavior was much more common before 2006 because these days you'd have to explain it in the CD&A. You had established goals and now you're varying from those goals—and with the CD&A those sorts of deviations receive a lot of scrutiny. Certainly the disclosures around those kinds of actions have caused companies to think twice. You'd need to have a really good reason” (R. Fox, personal communication, 2016).¹³

Jointly, these facts point towards the CD&A enhancing the viability of revenue-based pay as a commitment device by improving the credibility of pay package disclosures.

While the CD&A setting provides many features that make it a useful quasi-experiment for testing the theory of strategic delegation, there are also a number of dimensions on which it differs from the “ideal experiment.” For one, the ideal experiment would entail a stark transition from mandated non-disclosure to mandated disclosure (or alternatively, a stark transition from cheap talk disclosures to perfectly credible disclosures). In reality, firms were already disclosing information about their executives' pay packages, and could voluntarily disclose explicit pay package details. Some firms may even have had mechanisms in place to make their pay package disclosures both salient and credible; for such a firm, the CD&A may not have been as much of a “shock.”

Furthermore, the ideal experiment would involve explicit, binding statements about compensation arrangements going forward. In contrast, the CD&A is predominantly a backwards-looking document, detailing how the most recent pay outcomes were determined. Firms are required to describe any material changes in compensation going forward, but there is no concrete precedent as to whether a change in metrics would require such a preemptive disclosure. Most firms opt to disclose any changes in the metrics, so the plans disclosed in the CD&A tend to provide a relatively clear signal about executives' ongoing incentives (R. Fox, personal communication, 2019).¹⁴

Both departures from the ideal experiment explain why I might not find evidence of strategic contracting. Indeed, if firms were already perfectly able to commit through their pay packages prior to the CD&A, then I should not expect to see any change around adoption. Conversely, if firms were quite unable to commit through their pay packages, even after the CD&A (e.g., due to the backwards-looking

¹³ Robbi Fox has been a Senior Advisor with Exequity since 2010. Before that, she was a Principal and Senior Compensation Consultant with Hewitt Associates for almost 25 years.

¹⁴ However, firms typically do not disclose explicit performance targets in advance.

nature of the disclosure requirements), then I should also not expect to see any change around adoption. Fortunately, my identification strategy only requires that the CD&A improved the viability of strategic delegation. It need not be the case that firms were completely unable to use executive pay packages as commitment devices prior to the CD&A (nor perfectly able to, afterwards). My empirical design only requires that the CD&A enhanced the saliency and/or credibility with which a CEO's incentives could be relayed to rivals.¹⁵ Thus, these departures from the ideal experiment are only problematic insofar as they reduce the power of my tests.

More problematic would be a departure from the ideal experiment that could induce spurious inferences. In the context of my tests, one such departure regards the timing of the CD&A rollout. The new CD&A disclosure rules were not finalized until August 2006. December year-end firms would not have seen the finalized rules until after their end-of-March deadlines for changing performance pay plans, in order to retain tax exemptions under Section 162(m).¹⁶ One might therefore question whether these firms would have been able to react to the new rules in 2006. While it is conceivable that firms might not have had the chance to react, a possibility I explore in Section 5.6, it is unlikely that sophisticated firms would have been unaware of the impending new rules. The new disclosure rules were publicly proposed on January 27, 2006, giving December year-end firms approximately two full months to react in advance of their Section 162(m) deadlines. Facing little opposition in the ~20,000 comment letters received, the CD&A rules were approved in August 2006 “substantially as proposed” back in January 2006 (Securities and Exchange Commission, 2006b, page 9).

One general concern when using regulatory interventions as instruments is the endogenous nature of the decision to regulate. In my setting, one important institutional fact bolsters the claim of plausible exogeneity: the regulator responsible for the CD&A was concerned about a lack of transparency in executive compensation, not about product market competition (Securities and Exchange Commission, 2006a,b). Its stated mission is to “protect investors, maintain fair, orderly, and efficient markets, and facilitate capital formation” (Securities and Exchange Commission, 2016), making it implausible that their decision to regulate was driven by relevant product market concerns.

4. Data, sample, and variables

In this section I describe the data sources and sample selection processes used in my study. I also explain the construction of the variables used in my analyses.

¹⁵ See Section IA1 of the Internet Appendix for a more formal discussion of the role the CD&A plays in my analysis.

¹⁶ Section 162(m) of the Internal Revenue Code places a \$1 million deductibility limit for compensation. Performance-based pay can be exempt from this limit if the performance goals are objective and determined within the first quarter of the year.

4.1. Data sources and sample selection

My sample comes from the intersection of Compustat's Annual and Quarterly Industrial Files and Incentive Lab. I examine a short window around CD&A adoption: fiscal years 2002–2009.¹⁷ I then discard firm-year observations for which Incentive Lab codes the metric disclosure as “Vague” (roughly 4% of the sample, predominately from the pre-CD&A period).¹⁸ I further discard firms for which no data are available prior to CD&A adoption. Following prior studies, I include only September through December year-end firms in my sample (e.g., Ferri et al., 2018; Gipper, 2021). This selection process yields a final sample of 3117 firm-year observations from 539 unique firms.

4.2. Variable construction

Below, I define the variables used in my analyses. Summary statistics, split by mode of competition, are presented in Table 1.

4.2.1. Executive incentives

I measure executive incentives using an indicator variable, *Revenue-Based Pay*, equal to one if the CEO has any compensation tied to absolute performance “Sales” objectives, according to Incentive Lab. I use an extensive margin measure because intensive margin measures (i.e., the weights placed on revenue-based pay) cannot be reliably constructed from Incentive Lab data. This data limitation biases against any results, because it prevents me from identifying changes in the use of revenue-based pay, among firms that have always used it.

I also examine “other-financial” and “non-financial” performance metrics which, like revenue, are often included as secondary supplements to a CEO's incentive plan. I consider a metric to be other-financial if Incentive Lab codes it as an “Accounting” metric (*metrictype*=“Accounting”), but neither an earnings metric nor a revenue metric.¹⁹ Examples of other-financial metrics include cash flow, operating ratio, and gross margin. I consider a metric to be non-financial if Incentive Lab codes it as an “Other” metric (*metrictype*=“Other”). Examples of non-financial metrics include customer satisfaction, employee retention, and workplace safety. Using these definitions, I construct the indicator variables *Other-Financial* and *Non-Financial*, which equal one for firm-years in which the CEO has any absolute objectives tied to other-financial or non-financial metrics, respectively. I also construct count variables *#Other-Financial* and *#Non-Financial*, which reflect the number of other-financial and non-financial metrics present in the CEO's pay plan. Due to the right-skewness

¹⁷ For all observations, the fiscal year is coded as the calendar year of the fiscal closing date.

¹⁸ The advantage of this design choice is that it helps to ensure that my results are driven by changes in the underlying contracts, rather than changes in the quality of the data. My results are not sensitive to this choice.

¹⁹ I consider a metric to be an earnings or revenue metric if the variable *metric* contains any of the following strings: “Sales,” “Earnings,” “EPS,” “EBIT,” “EBITDA,” “Operating Income,” “ROA,” “ROI,” “ROE,” or “ROIC.”

Table 1

Summary statistics.

This table presents summary statistics for all variables used in my analyses. This sample consists of 3117 firm-year observations from 539 unique firms.

	Cournot Sample Obs. = 1728			Bertrand Sample Obs. = 1389		
	Mean	Med.	SD	Mean	Med.	SD
Executive Incentives						
Revenue-Based Pay	0.197	0.000	0.398	0.340	0.000	0.474
Other-Financial	0.496	0.000	0.500	0.379	0.000	0.485
Non-Financial	0.582	1.000	0.493	0.534	1.000	0.499
#Other-Financial	0.958	0.000	1.548	0.772	0.000	1.652
#Non-Financial	1.421	1.000	2.625	1.136	1.000	1.882
CD&A Adoption						
PostCD&A	0.534	1.000	0.499	0.541	1.000	0.499
Post2006	0.546	1.000	0.498	0.560	1.000	0.497
Post2007	0.402	0.000	0.490	0.413	0.000	0.492
Early Adopter	0.912	1.000	0.283	0.859	1.000	0.348
Market Share						
Share	0.135	0.046	0.199	0.166	0.075	0.217
ln(Share)	−3.183	−3.094	1.738	−3.075	−2.596	2.122
Firm and Industry Size						
ln(Sales)	7.771	7.866	1.458	7.702	7.746	1.636
ln(SIC Sales)	10.953	10.839	1.554	10.777	10.624	1.663
Product Market						
ln(HHI)	−1.984	−1.892	1.023	−1.772	−1.807	0.859
ln(COGS)	7.591	7.751	1.485	7.243	7.395	1.727
ln(Margin)	−0.780	−0.837	0.945	−0.346	−0.369	1.106
ln(Markup)	−1.822	−1.748	0.720	−1.859	−1.785	0.706

of the count measures, I add one and take their natural logarithms.

4.2.2. CD&A adoption

I exploit the mandated introduction of the CD&A as a shock to pay package disclosure. In my empirical analyses, I use *PostCD&A*, which takes a value of one during and after fiscal 2006 (2007) for December (September through November) year-end firms. In staggered adoption tests, I also construct the indicator variables *Post2006* (*Post2007*), which takes a value of one during and after fiscal 2006 (2007), and *Early Adopter*, which takes a value of one for December year-end firms.

4.2.3. Market share

I measure product market influence with market shares, defined based on firms' primary 4-digit SIC, according to Compustat. Because market shares are endogenous to changes in managerial incentives, I use a static firm-level measure, *Share*, equal to a firm's average pre-CD&A market share (starting in 2002). Intuitively, a firm that commands very little market share (either because it is small, or because the number of competitors is large) will not be able to manipulate rivals through strategic commitment. In contrast, a firm with a large market share (either because it is large, or because the number of competitors is small) stands to gain much more from strategic commitment. In my main regression analyses, I use the log-transformed measure, *ln(Share)*, to address the right-skewed distribution of the market shares.

4.2.4. Firm and industry size

I measure firm and industry size using GAAP revenues: *Sales* is a firm's average pre-CD&A revenue, and *SIC Sales*

is the sum of the average pre-CD&A GAAP revenues from all Compustat firms in a given 4-digit SIC. I use the natural logarithms to address right-skewness. Note that *Sales*, *SIC Sales*, and *Share* are all related by the identity: $\ln(\text{Share}) = \ln(\text{Sales}) - \ln(\text{SIC Sales})$.

4.2.5. Competitive mode

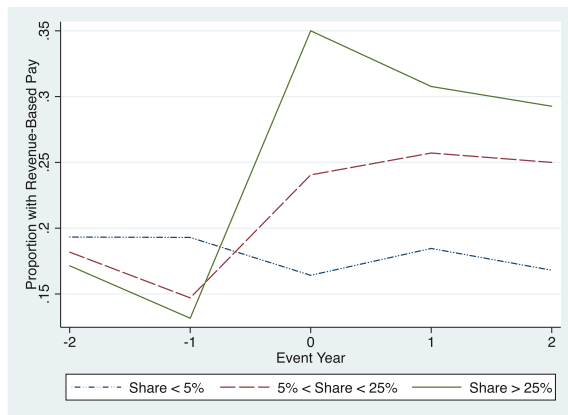
Whether a firm would benefit from a commitment to more aggressive or more passive behavior depends on whether strategic actions are substitutes (e.g., Cournot) or complements (e.g., Bertrand). My primary measure is based on the regression approach developed by Kedia (2006), designed to directly estimate the mode of competition from realized performance outcomes. The measure is constructed as follows. For each firm-year, I run the estimating equation:

$$\Delta \frac{\Delta \pi_{i,t}}{\Delta x_{i,t}} = \beta_1 x_{i,t} \Delta x_{i,t} + \beta_2 \Delta x_{i,t} + \beta_3 x_{i,t} \Delta x_{j,t} + \beta_4 \Delta x_{j,t}, \quad (1)$$

where π_i is firm i 's quarterly profits, x_i is firm i 's quarterly revenues, and x_j is the average contemporaneous quarterly revenue of all of firm i 's rivals (defined at the 4-digit SIC level). The sign of $\beta_3 x_i + \beta_4$ is the estimator of the sign of the strategic interaction, which reflects the mode of competition. Aggregating over all firm-years in a Fama-French 48 industry provides an industry-level measure of the competitive mode; I code industries as Cournot if the median sign is negative, and Bertrand if the median sign is positive. I refer to this measure as "Kedia."

Intuitively, the left-hand side variable is intended to capture the returns to more aggressive actions. With this interpretation in mind, $\beta_3 x_i + \beta_4$ reflects the extent to

Panel A: Cournot sample



Panel B: Bertrand sample

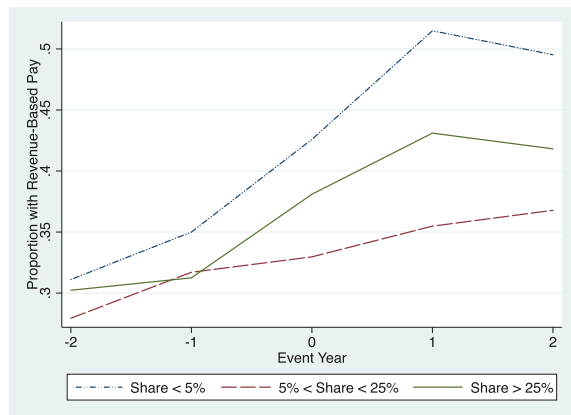


Fig. 1. Time series of revenue-based pay by market share. In this figure, I plot the time series of revenue-based pay in CEO pay packages, aligned in event time around CD&A adoption. I split the sample into three groups based on pre-CD&A market shares: $0\% \leq \text{Share} < 5\%$; $5\% < \text{Share} < 25\%$; and $\text{Share} > 25\%$. Panel A presents the time series for Cournot industries; Panel B presents the time series for Bertrand industries. Each line represents the proportion of CEOs who have incentive pay tied to absolute revenue-based performance goals. Year 0 represents each firms' adoption of the CD&A requirements.

which these marginal returns vary with rival aggressiveness (proxied for by x_j). If $\beta_3 x_i + \hat{\beta}_4 > 0$, then rivals' aggressiveness is estimated to increase the gains to aggressiveness (i.e., strategic actions are complements), and competition is said to be Bertrand. If $\beta_3 x_i + \hat{\beta}_4 < 0$, then rivals' aggressiveness is estimated to decrease the gains to aggressiveness (i.e., strategic actions are substitutes), and competition is said to be Cournot.

While quarterly sales are undoubtedly an imperfect proxy for aggressiveness (e.g., due to supply/demand shocks and/or other firms' actions affecting own-firm sales), the measure can be shown to be an unbiased estimator of competition type if marginal costs are flat and demand is linear. Moreover, under these assumptions, the measure allows for arbitrary shocks to supply and demand. In short, the measure was constructed in light of Bresnahan (1989) and Nevo's (1998) critiques, and will not naively classify industries as Cournot (Bertrand) simply because of negative (positive) correlation in rivals' performance outcomes (e.g., due to correlated supply and/or demand shocks). Instead, it estimates the sign of the cross-partial of marginal profits with respect to both own aggressiveness and rival firm aggressiveness.

This measure is conceptually ideal for two reasons: (1) it is based on the sign of the slopes of firms' best response functions, which dictates whether firms would prefer to commit to more aggressive or more passive behavior; and (2) it is flexible enough to reflect the sign of the strategic interdependencies across a wide array of possible games.

While this measure has appealing theoretical properties, estimating the mode of competition is notoriously difficult and no single measure is likely to capture it perfectly. For this reason, I supplement the primary measure with seven alternative measurement approaches. In the Appendix, I discuss the theoretical underpinning and empirical construction for each alternative measure and tabulate industry codings across all eight measures (Table A.1).

4.2.6. Product market

I measure product market concentration using the Herfindahl–Hirschman Index. I calculate the sum of the squared pre-CD&A average market shares and take the natural logarithm to construct the variable $\ln(HHI)$. I further examine three key product market outcomes intended to reflect firms' production quantities and markups. There is no broad sample data on firms' production quantities and product prices, so I rely on reported financial outcomes. One important caveat is that reported financial outcomes conflate quantities with per-unit prices and costs, which makes it difficult to cleanly assess changes in prices and quantities, separately.

I use the logarithm of cost of goods sold ("COGS") to measure output. I use two measures of margins/markups: gross profit scaled by COGS (*Margin*) and EBITDA scaled by sales (*Markup*). Due to the right-skewness of these measures, I use their natural logarithms in my analyses.

5. Empirical analysis

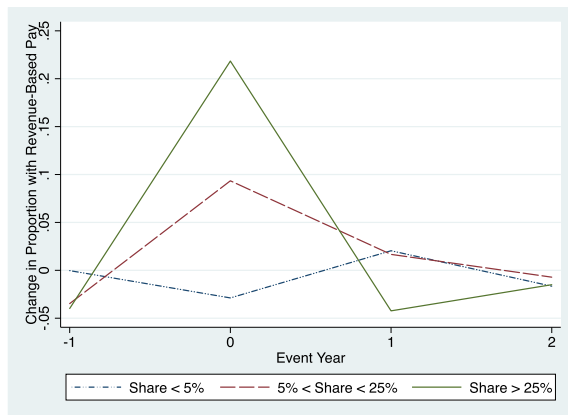
In this section, I describe my empirical methodology and present the findings.

5.1. Descriptive evidence

The first step in my empirical analysis is to document whether the use of revenue-based pay increased differentially after the CD&A for large-share firms in Cournot industries. I begin with a graphical examination of the time series trends in revenue-based pay. I split firms into three groups by market share: small-share ($\text{Share} < 5\%$); medium-share ($5\% < \text{Share} < 25\%$); and large-share ($25\% \leq \text{Share}$).

In Fig. 1, I plot the time series of revenue-based pay usage rates around CD&A adoption for each mode of competition (Cournot vs. Bertrand). Panel A (Panel B) depicts the time series in Cournot (Bertrand) industries. In Cournot industries, the small-share group shows no significant changes at all, while the medium-share group shows

Panel A: Cournot sample



Panel B: Bertrand sample

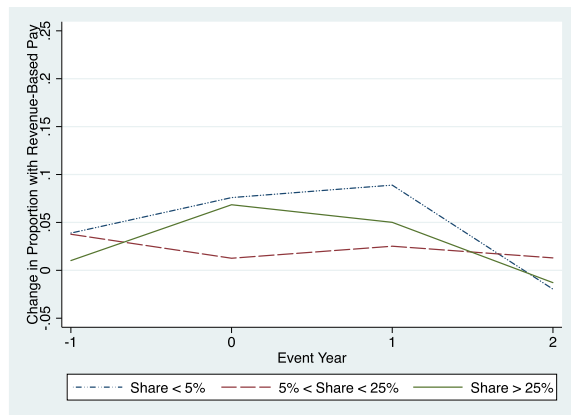


Fig. 2. Time series of changes in revenue-based pay by market share. In this figure, I plot the time series of changes in revenue-based pay in CEO pay packages, aligned in event time around CD&A adoption. I split the sample into three groups based on pre-CD&A market shares: $0\% \leq \text{Share} < 5\%$; $5\% < \text{Share} < 25\%$; and $\text{Share} > 25\%$. Panel A presents the time series for Cournot industries; Panel B presents the time series for Bertrand industries. Each line represents the year-over-year change in the proportion of CEOs who have incentive pay tied to absolute revenue-based performance goals. Year 0 represents each firms' adoption of the CD&A requirements.

a moderate uptick (from $\sim 15\%$ to $\sim 25\%$) in the first year of CD&A compliance and the large-share group shows a large uptick (from $\sim 15\%$ to $\sim 35\%$) in the first year of CD&A compliance. Among Bertrand firms, there is a secular upward trend over the sample period (both before and after CD&A adoption), with no obvious differences across small-, medium- and large-share groups, nor any unusual changes around the CD&A adoption year.

I next augment the graphical presentation using a first-differences design. In Fig. 2, I plot the year-over-year changes in the proportion of firms using revenue-based pay, split as before by market share and strategic mode. In Cournot industries, there is a notable spike in the rate of revenue-based pay adoption in first year of CD&A compliance, but only among medium- and large-share firms. In Bertrand industries, the rate of revenue-based pay adoption remains fairly stable with no notable changes for any group. Collectively, Figs. 1 and 2 depict a consistent story. In Cournot industries, medium- and especially large-share firms appear to react to the CD&A by incorporating revenue-based pay into their CEOs' incentive plans, while small-share firms do not. In Bertrand industries, there is a consistent upward trend in the use of revenue-based pay, with no obvious reaction to the CD&A at all.

Table 2 provides descriptive statistics on the changes in revenue-based pay around CD&A adoption using a finer partition of firms into market share bins. I split firms into six bins by market share: 0–1%, 1–5%, 5–10%, 10–25%, 25–65%, and 65–100%, and report the average utilization rates of revenue-based pay. Panel A presents the utilization rates in the last year prior to CD&A adoption while Panel B presents the utilization rates in first year after CD&A adoption and Panel C presents the change in utilization rates from pre to post.

Among Cournot firms, the change in the use of revenue-based pay exhibits a stark inverted U-shaped pattern. The use of revenue-based pay increases substantially among

the firms with 10–25% market shares and by even more among the firms with 25–65% market shares. These are the firms for which the strategic value of revenue-based pay is likely to be greatest. Among more atomistic firms and more monopolistic firms (for which the strategic benefits of revenue-based pay are negligible), I observe no shift towards revenue-based pay. These results are all closely aligned with the predictions from Fershtman and Judd (1987).

Among Bertrand firms, the change in the use of revenue-based pay exhibits a non-inverted U-shaped pattern. The largest increases in revenue-based pay usage occur among the most atomistic and monopolistic firms (i.e., those with shares less than 1% or greater than 65%). These are precisely the firms for which the addition of revenue-based pay would be least costly, strategically. More oligopolistic Bertrand firms (e.g., those with market shares between 10% and 65%) exhibit a much smaller shift towards revenue-based pay. I caution that these results are not as easily interpretable given the underlying time trend present among these firms. The pre- versus post-CD&A changes do not necessarily reflect firms' responses to the CD&A, rather they appear to be part of a gradual time trend towards revenue-based pay, as shown in Fig. 1B.

Strategic considerations can explain the inverted and non-inverted U-shaped patterns in Panel C, but cannot justify Bertrand firms' overall shift towards revenue-based pay. From a purely strategic perspective, these firms should avoid using revenue-based pay. One likely explanation for these patterns is that the agency-theoretic importance of revenue-based pay increased for Bertrand firms over my sample period, and the strategic costs of revenue-based pay represent a minor impediment to its employ. Atomistic and monopolistic firms do not bear these costs as much and are therefore most likely to incorporate revenue-based incentives. Oligopolists bear these costs much more heavily and are therefore less likely to adopt revenue-based pay.

Table 2

Firms' use of revenue-based pay, by market share.

This table provides descriptive statistics for firms' use of revenue-based pay, split into six categories by market share: 0%–1%, 1%–5%, 5%–10%, 10%–25%, 25%–65%, and 65%–100%. Panel A presents statistics for firms' final year before CD&A compliance; Panel B presents statistics for the first year of CD&A compliance; and Panel C presents the changes from pre- to post-CD&A. Within each panel, statistics are split by mode of competition, where # of Firms is the number of firms in each market share category and Proportion is the fraction of those firms using revenue-based pay in the CEO's pay package.

Panel A: Final year pre-CD&A					
	Cournot Sample		Bertrand Sample		
	# of Firms	Proportion	# of Firms	Proportion	
0%<Share<1%	70	0.243	62	0.355	
1%<Share<5%	84	0.167	41	0.268	
5%<Share<10%	36	0.139	33	0.182	
10%<Share<25%	50	0.200	58	0.345	
25%<Share<65%	42	0.190	41	0.293	
65%<Share≤100%	11	0.182	11	0.273	
Panel B: First year post-CD&A					
	Cournot Sample		Bertrand Sample		
	# of Firms	Proportion	# of Firms	Proportion	
0%<Share<1%	70	0.200	62	0.484	
1%<Share<5%	84	0.143	41	0.317	
5%<Share<10%	36	0.167	33	0.242	
10%<Share<25%	50	0.280	58	0.379	
25%<Share<65%	42	0.429	41	0.317	
65%<Share≤100%	11	0.182	11	0.455	
Panel C: Change from pre- to post-CD&A					
	Cournot Sample		Bertrand Sample		
	# of Firms	ΔProportion	# of Firms	ΔProportion	
0%<Share<1%	70	−0.043	62	0.129	
1%<Share<5%	84	−0.024	41	0.049	
5%<Share<10%	36	0.028	33	0.061	
10%<Share<25%	50	0.080	58	0.034	
25%<Share<65%	42	0.238	41	0.024	
65%<Share≤100%	11	0.000	11	0.182	

Analyzing the non-strategic rationale for revenue-based pay lies beyond the intended scope of this study. As noted in Section 2, the purpose of this study is to assess whether strategic considerations are incrementally important vis-à-vis firms' executive compensation practices. As such, I focus on isolating the variation in pay practices for which strategic considerations are most likely responsible.

5.2. Empirical approach

My empirical approach is as follows. Conceptually, I examine how firms' use of revenue-based pay responds to the interactive effects of three economic forces: (1) the magnitude of the benefits of strategic commitment (captured by market shares); (2) the viability of using pay packages for the purpose of strategic commitment (captured by CD&A adoption); and (3) the preferred direction of strategic commitment (captured by competitive mode). Framed in this context, my key prediction is as follows: the relation between market shares and revenue-based pay will become more positive after CD&A adoption, specifically in Cournot industries.

In essence, I combine two continuous treatment difference-in-differences estimators (one for Cournot industries and one for Bertrand industries) to form a

triple-differences estimator. One advantage of the triple-differences design is that it relies on a very lenient identifying assumption: absent the introduction of the CD&A, the slope of the relation between revenue-based pay and market shares would have been parallel across Cournot and Bertrand industries. Note that this design is not compromised by average level differences in the use of revenue-based pay across Cournot and Bertrand industries, nor by differential time trends in the use of revenue-based pay across Cournot and Bertrand industries.

While the identifying assumption cannot be directly tested, it is possible to assess the plausibility of the parallel trends assumption graphically. To do so, I estimate the following regression for each year and mode of competition:

$$\text{Revenue-Based Pay}_{i,t} = \alpha_t + \beta_t \ln(\text{Share}_i) + \varepsilon_{i,t}. \quad (2)$$

This procedure yields two estimates of β_t for each year t ; one estimate for Cournot industries and one for Bertrand industries, which I present in Fig. 3. For ease of comparison, I normalize each time series to start at zero by subtracting $\hat{\beta}_{2002}$ from each $\hat{\beta}_t$. In Panel A, I plot the coefficient estimates for Cournot and Bertrand firms; in Panel B, I plot the differences between Cournot and Bertrand coefficient estimates, along with their 95% confidence intervals.

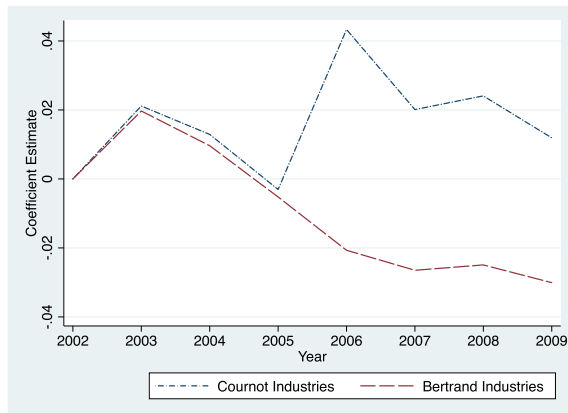
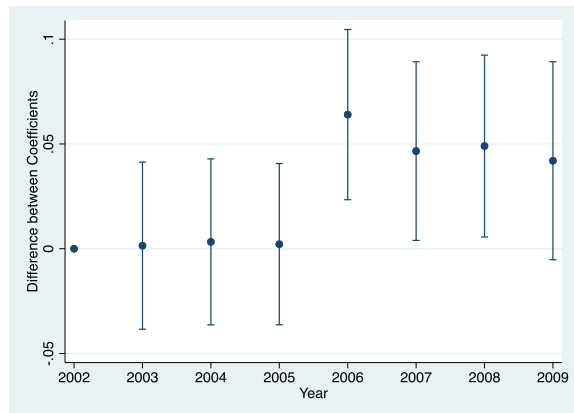
Panel A: Coefficients on $\ln(\text{Share})$ Panel B: Differences between coefficients on $\ln(\text{Share})$ 

Fig. 3. Time series of the relation between revenue-based pay and market share, by mode of competition. In this figure, I plot the slope of the relation between *Revenue-Based Pay* and $\ln(\text{Share})$, by year, split by the mode of competition (Cournot versus Bertrand). For each year, and both modes of competition, the slope of the relation is estimated as β^i from the following regression specification:

$$\text{Revenue-Based Pay}_{i,t} = \alpha_t + \beta_i \ln(\text{Share}_i) + \varepsilon_{i,t},$$

where *Share* is a firm's average pre-CD&A market share and *Revenue-Based Pay* is an indicator variable equal to one if the CEO has absolute incentive pay tied to revenue. Panel A shows the time series of β_i 's for both modes of competition, while Panel B shows the time series of the difference between them, along with the 95% confidence interval of the difference. Each time series is normalized to start at zero.

The assumption that β_i s would have followed parallel trends appears reasonable. In the pre-CD&A period, the slope coefficients move almost in lockstep. When the CD&A takes effect, the slope coefficients diverge to become more positive for Cournot industries. After the shock, slope coefficients remain apart but resume approximately parallel motion.

5.3. Main analysis

I test for the predicted interactive effect using the following regression:

$$\text{Revenue-Based Pay}_{i,t} = \beta \ln(\text{Share}_i) \times \text{PostCD\&A}_t + \mu_i + \tau_t + \varepsilon_{i,t}, \quad (3)$$

where *Share* is a firm's pre-CD&A average market share, *PostCD&A* is an indicator variable, taking a value of one if a firm has adopted the CD&A, and μ and τ are firm and year fixed effects, respectively. In all of my regression analyses, each 12-month period receives a unique fixed effect. That is, the years for the year fixed effects are defined based on the combination of fiscal year and fiscal year-end month.²⁰ This design choice ensures that all ostensibly within-year variation actually comes from the same 12-month time period. This fixed effect structure subsumes the main effects of $\ln(\text{Share})$ and *PostCD&A*. I run the regression separately for Cournot and Bertrand industries and test for differences across specifications.

The results, tabulated in Table 3, align closely with my predictions; $\hat{\beta}$ is both positive and significant for the Cournot sample, suggesting that the firms that stand to gain more from strategic commitment are more likely to add revenue-based pay after the CD&A. In contrast, $\hat{\beta}$ is

Table 3

Revenue-based pay by market share and CD&A adoption.

This table presents evidence on the association between market share and revenue-based incentives in CEO pay packages, before versus after the CD&A. Each specification is based on the regression:

$$\text{Revenue-Based Pay}_{i,t} = \beta \ln(\text{Share}_i) \times \text{PostCD\&A}_t + \mu_i + \tau_t + \varepsilon_{i,t},$$

where *Share* is the firm's pre-CD&A market share (based on primary 4-digit SIC), *PostCD&A* is an indicator variable equal to one for firm-years after mandated CD&A compliance, and μ and τ are firm and year fixed effects, respectively. The dependent variable is an indicator equal to one for firm-years in which the CEO has compensation tied to absolute revenue objectives. Specifications differ only with respect to the sample. Specification 1 presents results for Cournot industries; Specification 2 presents results for Bertrand industries. Column 3 presents the results of a test of the differences in coefficients between Specifications 1 and 2. Below each coefficient estimate, I report a *t*-statistic calculated using robust standard errors clustered by firm. *, **, and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. All significance levels are based on two-tailed tests.

Variables	Outcome = <i>Revenue-Based Pay</i>		
	(1)	(2)	(1) – (2)
$\ln(\text{Share}) \times \text{PostCD\&A}$	0.034*** (2.754)	–0.016 (–1.308)	0.050*** (2.881)
Competition Type	Cournot	Bertrand	
Firm Fixed Effects	Yes	Yes	
Year Fixed Effects	Yes	Yes	
Observations	1728	1389	
R-squared	0.694	0.707	

negative but insignificant among Bertrand firms. The difference in coefficients between the Cournot and Bertrand specifications is significant as well. In terms of economic magnitudes, the results indicate that a one standard deviation increase in $\ln(\text{Share})$ corresponds to a roughly 6 percentage point increase in the probability that a Cournot

²⁰ For example, fiscal years ending in December 2005 get one fixed effect, while fiscal years ending in November 2005 get a different one.

firm will adopt revenue-based pay in the post-CD&A period.

5.4. Refinements

The preceding analysis establishes that market shares and revenue-based pay become more positively associated in Cournot industries, but not in Bertrand industries. In this subsection, I offer several refinements of the main analysis to better establish the strategic impetus underlying these patterns. First, I leverage the theoretical distinction between market share and firm size. Second, I exploit the staggered nature of the CD&A roll-out. Third, I replicate the main analysis on a matched sample of Cournot and Bertrand firms.

5.4.1. Firm size versus market share

Whether a firm benefits from strategic commitment has much more to do with the firm's market share than its size. To illustrate, consider a large firm that competes as one of many large players in a competitive multi-billion dollar industry; such a firm has little market share and will not benefit from commitment as its actions do not have much bearing on the market. In contrast, consider a small firm that commands a large portion of a niche industry; such a firm will have much to gain from commitment as it wields considerable power to affect the market in which it competes.

As an empirical matter, firm size and market share are closely related; in my sample, firm size and market share have a correlation of around 0.6. Thus, one might be concerned that the results in Table 3 can be attributed to a firm size effect that is unrelated to strategic considerations; perhaps larger firms react differentially to the CD&A due to heterogeneous sophistication and/or regulatory attention. To allay this concern, I offer two sets of additional tests, which help establish that the previously documented patterns are best attributed to market share, not firm size.

First, I augment the regression specification in Eq. (3) by adding a control for firm size:

$$\text{Revenue-Based Pay}_{i,t} = \beta_1 \ln(\text{Share}_i) \times \text{PostCD\&A}_t + \beta_2 \ln(\text{Sales}_i) \times \text{PostCD\&A}_t + \mu_i + \tau_t + \varepsilon_{i,t}, \quad (4)$$

where *Sales* is a firm's pre-CD&A average revenue. If the results in Table 3 are driven by a size effect, then a control for $\ln(\text{Sales})$ should steal the loading from $\ln(\text{Share})$. This is not what I find in the data, as shown in Table 4, Specifications 1 and 2. In Cournot industries, the estimated coefficient on $\ln(\text{Share}) \times \text{PostCD\&A}$ increases slightly (from 0.034 to 0.035), and the estimated slope coefficient on $\ln(\text{Sales}) \times \text{PostCD\&A}$ is statistically insignificant and economically minuscule. Neither variable loads significantly for Bertrand industries, and the difference between specifications is significant only for market share.

Second, I exploit the following identity:

$$\ln(\text{Share}) = \ln(\text{Sales}) - \ln(\text{SIC Sales}), \quad (5)$$

where *SIC Sales* is the pre-CD&A average of 4-digit SIC-level revenue. Using this identity, I alter the main regression specification by replacing $\ln(\text{Share})$ with its two com-

ponent parts:

$$\begin{aligned} \text{Revenue-Based Pay}_{i,t} = & \beta_1 \ln(\text{Sales}_i) \times \text{PostCD\&A}_t \\ & + \beta_2 \ln(\text{SIC Sales}_i) \times \text{PostCD\&A}_t + \mu_i + \tau_t + \varepsilon_{i,t}. \end{aligned} \quad (6)$$

If the results in Table 3 are driven by a size effect, then $\ln(\text{SIC Sales}) \times \text{PostCD\&A}$ should not load. If the results in Table 3 are driven by a market share effect, then $\ln(\text{Sales}) \times \text{PostCD\&A}$ and $\ln(\text{SIC Sales}) \times \text{PostCD\&A}$ should both load with similar magnitudes and opposite signs. I find the evidence in favor of the latter, as shown in Table 4, specifications 3 and 4. In Cournot industries, $\ln(\text{Sales}) \times \text{PostCD\&A}$ and $\ln(\text{SIC Sales}) \times \text{PostCD\&A}$ load with opposite signs but with similar magnitudes (0.032 vs. -0.035). Neither variable carries a significant loading in Bertrand industries, and the difference across specifications is significant for both coefficients.

These results establish that the changes in revenue-based pay are predominantly related to market shares. Notably, the negative and significant loading on $\ln(\text{SIC Sales}) \times \text{PostCD\&A}$ bolsters the "strategic commitment" interpretation, as industry size plays a first-order role in shaping the strategic landscape but is typically not considered important outside of strategic contexts. Put bluntly, why should industry size matter (incrementally to firm size) other than through its effect on strategic considerations?

5.4.2. Staggered adoption

Not all firms were treated by the CD&A simultaneously. December year-end firms were required to comply for fiscal 2006, while September through November year-end firms were not required to comply until fiscal 2007. If the previously documented patterns are truly caused by the CD&A (as opposed to some coincident factor), then the effect should manifest for December year-end firms before the rest.

I test for the effect of the CD&A on revenue-based pay using the following regression specification:

$$\begin{aligned} \text{Revenue-Based Pay}_{i,t} = & \beta_1 \text{Early Adopter}_i \times \ln(\text{Share}_i) \times \text{Post2006}_t \\ & + \beta_2 \text{Early Adopter}_i \times \ln(\text{Share}_i) \times \text{Post2007}_t \\ & + \beta_3 \text{Early Adopter}_i \times \ln(\text{Sales}_i) \times \text{Post2006}_t \\ & + \beta_4 \text{Early Adopter}_i \times \ln(\text{Sales}_i) \times \text{Post2007}_t \\ & + \beta_5 \ln(\text{Share}_i) \times \text{Post2006}_t + \beta_6 \ln(\text{Share}_i) \\ & \times \text{Post2007}_t \\ & + \beta_7 \ln(\text{Sales}_i) \times \text{Post2006}_t + \beta_8 \ln(\text{Sales}_i) \\ & \times \text{Post2007}_t + \mu_i + \tau_t + \varepsilon_{i,t}, \end{aligned} \quad (7)$$

where *Post2006* (*Post2007*) is an indicator variable, equal to one during and after 2006 (2007) and *Early Adopter* is an indicator variable equal to one for December year-end firms. I restrict the sample to the three-year window from 2005 to 2007. This window contains one year where no firms are treated (2005), one year where all firms are treated (2007), and one year in between where only December year-end firms are treated (2006). The coefficients of interest are β_1 and β_2 , which reflect the effect of adoption year on the timing of firms' shifts towards revenue-based pay. As in the prior analyses, I run the regression

Table 4

Market share, firm size and market size.

This table presents evidence on the effects of market share, firm size, and market size on post-CD&A shifts towards revenue-based pay. Specifications 1 and 2 exactly replicate the analysis in Table 3, but include $\ln(\text{Sales}) \times \text{PostCD\&A}$ as an additional regressor. Specifications 3 and 4 replace $\ln(\text{Share})$ with its two component parts, $\ln(\text{Sales})$ and $\ln(\text{SIC Sales})$. Odd-numbered specifications present results for Cournot industries; even-numbered specifications present results for Bertrand industries. After each pair of specifications, I present a test of the differences in coefficients between Cournot and Bertrand samples. Below each coefficient estimate, I report a *t*-statistic calculated using robust standard errors clustered by firm. *, **, and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. All significance levels are based on two-tailed tests.

Variables	Outcome = Revenue-Based Pay					
	(1)	(2)	(1) – (2)	(3)	(4)	(3) – (4)
$\ln(\text{Share}) \times \text{PostCD\&A}$	0.035*** (2.818)	–0.005 (–0.348)	0.040** (2.097)			
$\ln(\text{Sales}) \times \text{PostCD\&A}$	–0.003 (–0.179)	–0.023 (–0.998)	–0.023 (–1.001)	0.032* (1.899)	–0.028 (–1.503)	0.060** (2.395)
$\ln(\text{SIC Sales}) \times \text{PostCD\&A}$				–0.035*** (–2.819)	0.005 (0.348)	–0.040** (–2.097)
Competition Type	Cournot	Bertrand		Cournot	Bertrand	
Firm Fixed Effects	Yes	Yes		Yes	Yes	
Year Fixed Effects	Yes	Yes		Yes	Yes	
Observations	1728	1389		1728	1389	
R-squared	0.694	0.708		0.694	0.708	

for Cournot and Bertrand industries separately and test for differences across specifications.

The results, presented in Table 5, show that variation in adoption dates explains variation in firms' use of revenue-based pay. In Cournot industries, large-share firms begin using revenue-based pay in their adoption years: December year-end firms in 2006 (as evidenced by the positive $\hat{\beta}_1$), and September through November year-end firms in 2007 (as evidenced by the negative $\hat{\beta}_2$). Moreover, the economic magnitudes of the two coefficients are similar (0.107 vs. 0.126), indicating that the wedge that arose in 2006, when only early adopters were treated, is offset in 2007 when the late adopters become treated.

In Bertrand industries, I observe no similar patterns. Neither $\hat{\beta}_1$ nor $\hat{\beta}_2$ can be statistically distinguished from zero, and they are roughly 3.5–8.9 times smaller in magnitude than the corresponding coefficient estimates from the Cournot industries specification. The difference between the specifications is statistically significant for $\hat{\beta}_1$. For $\hat{\beta}_2$, the estimated difference is economically large but not statistically significant. Perhaps the lack of significance can be attributed to the fact that late adopters (i.e., firms with September through November fiscal year-ends) comprise less than 12% of the sample.

As a caveat, fiscal year-ends are not assigned at random; firms with December versus non-December year-ends are systematically different from each other. To the extent that such differences are constant over my sample period, firm fixed effects will eliminate the potential source of bias. However, I cannot rule out the possibility that firms with different year-ends react differentially to omitted factors.

5.4.3. Coarsened exact matching

In the preceding analysis, I compare all Cournot firms in my sample to all Bertrand firms in my sample. Modes of strategic conduct are highly endogenous, so the com-

parison is not apples to apples. While I cannot exogenously shock an industry's mode of competition, I can ensure that my treatment and control samples are more closely comparable with respect to relevant observable characteristics. To this end, I use coarsened exact matching to match Cournot firm-year observations to Bertrand firm-year observations that are approximately identical with respect to known determinants of revenue-based pay: advertising intensity, R&D intensity, and firm age (Bloomfield et al., 2020). I also match on market share, given its central role in my research design.

The matching procedure is as follows. First, I coarsen each of the four match variables, by year, into quintiles or deciles. Second, I count the number of Cournot and Bertrand observations, by year, for each combination of coarsened characteristics. Third, I weight each firm-year observation based on these counts: Cournot observations for which at least one exact Bertrand match exists get a weight of 1; Bertrand observations get weighted so as to perfectly counter-balance the Cournot firms to which they are matched.²¹ Observations are dropped if they do not have an exact match in the same year from the opposite mode of competition.

I replicate the main analysis on the matched sample and present the results in Table 6. In Specifications 1 and 2 (Specifications 3 and 4), I present results for the quintile-matched (decile-matched) samples. Across both matching procedures, the estimated coefficient is positive and significant among Cournot industries and negative but insignificant among Bertrand industries. The difference between the two coefficients is also significant.

²¹ For example, suppose that in 2008 there are ten firms in the lowest quintile on all four characteristics. If three of these ten firms are Cournot, then those three observations get included in the matched sample with a weight of 1, each. The seven matched Bertrand observations would be included in the sample with a weight of 3/7th's each.

Table 5

Staggered adoption.

This table provides evidence that heterogeneity in CD&A adoption dates explains variation in revenue-based pay trends. Each specification is based on the following regression:

$$\begin{aligned} \text{Revenue-Based Pay}_{i,t} = & \beta_1 \text{Early Adopter}_i \times \ln(\text{Share}_i) \times \text{Post2006}_t + \beta_2 \text{Early Adopter}_i \times \ln(\text{Share}_i) \times \text{Post2007}_t \\ & + \beta_3 \text{Early Adopter}_i \times \ln(\text{Sales}_i) \times \text{Post2006}_t + \beta_4 \text{Early Adopter}_i \times \ln(\text{Sales}_i) \times \text{Post2007}_t \\ & + \beta_5 \ln(\text{Share}_i) \times \text{Post2006}_t + \beta_6 \ln(\text{Share}_i) \times \text{Post2007}_t \\ & + \beta_7 \ln(\text{Sales}_i) \times \text{Post2006}_t + \beta_8 \ln(\text{Sales}_i) \times \text{Post2007}_t + \mu_i + \tau_t + \varepsilon_{i,t}, \end{aligned}$$

where *Post2006* (*Post2007*) is an indicator variable, equal to one during and after 2006 (2007), *Early Adopter* is an indicator variable equal to one for December year-end firms, and μ and τ are firm and year fixed effects, respectively. The dependent variable is an indicator equal to one for firm-years in which the CEO has compensation tied to absolute revenue objectives. The sample is restricted to include only the years 2005, 2006, and 2007. Specifications differ only with respect to the sample. Specification 1 presents results for Cournot industries; Specification 2 presents results for Bertrand industries. Column 3 presents the results of a test of the differences in coefficients between Specifications 1 and 2. Below each coefficient estimate, I report a *t*-statistic calculated using robust standard errors clustered by firm. *, **, and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. All significance levels are based on two-tailed tests.

Variables	Outcome = Revenue-Based Pay		
	(1)	(2)	(1) – (2)
<i>ln(Share) × Post2006 × Early Adopter</i>	0.107** (2.083)	–0.012 (–0.385)	0.119** (1.985)
<i>ln(Share) × Post2007 × Early Adopter</i>	–0.126** (–1.991)	–0.036 (–0.669)	–0.090 (–1.082)
<i>ln(Sales) × Post2006 × Early Adopter</i>	–0.122 (–1.604)	0.008 (0.201)	–0.131 (–1.499)
<i>ln(Sales) × Post2007 × Early Adopter</i>	0.121 (1.497)	0.056 (0.683)	0.064 (0.556)
<i>ln(Share) × Post2006</i>	–0.063 (–1.289)	0.012 (0.475)	–0.074 (–1.362)
<i>ln(Share) × Post2007</i>	0.098 (1.571)	0.043 (0.816)	0.055 (0.676)
<i>ln(Sales) × Post2006</i>	0.123* (1.667)	–0.027 (–0.851)	0.150* (1.865)
<i>ln(Sales) × Post2007</i>	–0.104 (–1.311)	–0.082 (–1.017)	–0.022 (–0.195)
Competition Type	Cournot	Bertrand	
Firm Fixed Effects	Yes	Yes	
Year Fixed Effects	Yes	Yes	
Observations	708	587	
R-squared	0.848	0.854	

In sum, it does not appear that my findings can be explained by these determinants of revenue-based pay. If anything, controlling for these characteristics makes the results sharper—the coefficient estimates are as large or larger than those in Table 3.

5.5. Supplemental analyses

5.5.1. Other performance metrics

In the first year after CD&A adoption, roughly two-thirds of the firms in my sample increase the number of metrics used for executive performance evaluation. The average (median) firm increases the number of metrics by about 100% (50%). By itself, the post-CD&A increase in the number of performance metrics is not a concern; in fact, it is support for the notion that the CD&A improved metric binding. However, if large-share Cournot firms were differentially likely to expand the set of metrics in response to the CD&A, then the documented patterns in revenue-based pay might reflect this secular expansion, rather than strategic considerations related specifically to

revenue-based pay. I address this possibility with two different approaches.

First, I use placebo analyses to examine whether similar results are obtained for other-financial or non-financial metrics. If the previously documented results are driven by a secular increase in metrics, then similar patterns would likely manifest among these categories. I examine this possibility using the following regression specification:

$$\langle \text{Metric} \rangle_{i,t} = \beta \ln(\text{Share}_i) \times \text{PostCD\&A}_t + \mu_i + \tau_t + \varepsilon_{i,t}, \quad (8)$$

where *Metric* is one of two possible outcome variables: *Other-Financial* or *Non-Financial*.

Table 7 presents the results. In Specifications 1 and 2, I examine other-financial metrics and find no evidence that large-share firms differentially changed their usage in either Cournot or Bertrand industries. The estimated coefficient is statistically insignificant and economically negligible in both specifications. The difference between specifications is also insignificant. In Specifications 3 and 4, I ex-

Table 6

Matched sample.

This table presents results from replications of Table 3 using matched samples of Cournot and Bertrand firms. Cournot firm-year observations are matched to Bertrand firm-year observations, from the same year, using coarsened exact matching. The matching variables are: R&D intensity, advertising intensity, firm age, and market share. Matched Cournot observations are given a weight of 1; matched Bertrand observations are given a weight so as to perfectly counterbalance the Cournot firms to which they are matched. In Specifications 1 and 2 (3 and 4), I coarsen each match variable into quintiles (deciles) before matching. Odd-numbered specifications present results for Cournot industries; even-numbered specifications present results for Bertrand industries. After each pair of specifications, I present a test of the differences in coefficients between Cournot and Bertrand samples. Below each coefficient estimate, I report a *t*-statistic calculated using robust standard errors clustered by firm. *, **, and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. All significance levels are based on two-tailed tests.

Variables	Outcome = Revenue-Based Pay					
	Match by Quintiles			Match by Deciles		
	(1)	(2)	(1) – (2)	(3)	(4)	(3) – (4)
$\ln(\text{Share}) \times \text{PostCD\&A}$	0.050*** (3.420)	–0.014 (–0.771)	0.064*** (2.721)	0.035* (1.698)	–0.042 (–1.500)	0.077** (2.225)
Competition Type	Cournot	Bertrand		Cournot	Bertrand	
Firm Fixed Effects	Yes	Yes		Yes	Yes	
Year Fixed Effects	Yes	Yes		Yes	Yes	
Observations	1094	710		406	287	
R-squared	0.668	0.776		0.670	0.756	

Table 7

Other types of performance metrics.

This table presents results from replications of Table 3 using different outcome variables. In Specifications 1 and 2, the dependent variable is *Other-Financial*, which is an indicator equal to one for firm-years in which the CEO has incentive compensation tied to absolute financial objectives other than earnings or sales. In Specifications 3 and 4, the dependent variable is *Non-Financial*, which is an indicator equal to one for firm-years in which the CEO has compensation tied to non-financial absolute objectives. Odd-numbered specifications present results for Cournot industries; even-numbered specifications present results for Bertrand industries. After each pair of specifications, I present a test of the differences in coefficients between Cournot and Bertrand samples. Below each coefficient estimate, I report a *t*-statistic calculated using robust standard errors clustered by firm. *, **, and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. All significance levels are based on two-tailed tests.

Variables	Outcome = Other-Financial			Outcome = Non-Financial		
	(1)	(2)	(1) – (2)	(3)	(4)	(3) – (4)
	(1)	(2)	(1) – (2)	(3)	(4)	(3) – (4)
$\ln(\text{Share}) \times \text{PostCD\&A}$	0.014 (0.849)	0.005 (0.397)	0.009 (0.455)	0.053*** (2.820)	0.045*** (3.414)	0.008 (0.348)
Competition Type	Cournot	Bertrand		Cournot	Bertrand	
Firm Fixed Effects	Yes	Yes		Yes	Yes	
Year Fixed Effects	Yes	Yes		Yes	Yes	
Observations	1728	1389		1728	1389	
R-squared	0.615	0.631		0.547	0.607	

amine non-financial metrics. Unlike other-financial metrics, I observe a differential increase in the use of non-financial metrics among large-share firms. However, this increase is similar across both Cournot and Bertrand industries, with the difference across specifications being both economically tiny and statistically insignificant.

Second, I re-examine firms' reliance on revenue-based pay after controlling for other-financial and non-financial metrics. I do so with the following pair of regression specifications:

$$\begin{aligned} \text{Revenue-Based Pay}_{i,t} = & \beta_1 \ln(\text{Share}_i) \times \text{PostCD\&A}_t \\ & + \beta_2 \text{Other-Financial}_{i,t} + \beta_3 \text{Non-Financial}_{i,t} \\ & + \mu_i + \tau_t + \varepsilon_{i,t}, \end{aligned} \quad (9)$$

$$\begin{aligned} \text{Revenue-Based Pay}_{i,t} = & \beta_1 \ln(\text{Share}_i) \times \text{PostCD\&A}_t \\ & + \beta_2 \ln(1 + \#\text{Other-Financial}_{i,t}) \\ & + \beta_3 \ln(1 + \#\text{Non-Financial}_{i,t}) + \mu_i + \tau_t + \varepsilon_{i,t}. \end{aligned} \quad (10)$$

Table 8 presents the results from both regressions. In Specifications 1 and 2 (Specifications 3 and 4), I present the results using the indicator (logged) control variables. I find that other-financial and non-financial metrics help explain firms' use of revenue-based pay. However, controlling for them does not alter my inferences. In sum, it does not appear that a secular increase in performance metrics can explain my findings.

Table 8

Controlling for secondary metrics.

This table presents results from replications of Table 3 after controlling for the presence of other-financial and non-financial metrics in CEOs' incentive plans. In Specifications 1 and 2, I use indicator variable controls: *Other-Financial* and *Non-Financial*. In Specifications 3 and 4, I use logged controls: $\ln(1 + \#Other-Financial)$ and $\ln(1 + \#Non-Financial)$. Odd-numbered specifications present results for Cournot industries; even-numbered specifications present results for Bertrand industries. After each pair of specifications, I present a test of the differences in coefficients between Cournot and Bertrand samples. Below each coefficient estimate, I report a *t*-statistic calculated using robust standard errors clustered by firm. *, **, and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. All significance levels are based on two-tailed tests.

Variables	Outcome = Revenue-Based Pay					
	(1)	(2)	(1) – (2)	(3)	(4)	(3) – (4)
$\ln(Share) \times PostCD\&A$	0.032** (2.590)	–0.019 (–1.600)	0.051*** (2.969)	0.033*** (2.773)	–0.020* (–1.691)	0.054*** (3.157)
<i>Other-Financial</i>	0.060** (2.225)	0.091* (1.896)	–0.031 (–0.567)			
<i>Non-Financial</i>	0.029 (1.096)	0.065** (2.039)	–0.036 (–0.865)			
$\ln(1 + \#Other-Financial)$				0.051** (2.258)	0.090** (2.187)	–0.039 (–0.828)
$\ln(1 + \#Non-Financial)$				0.014 (0.564)	0.065** (2.419)	–0.051 (–1.423)
Competition Type	Cournot	Bertrand		Cournot	Bertrand	
Firm Fixed Effects	Yes	Yes		Yes	Yes	
Year Fixed Effects	Yes	Yes		Yes	Yes	
Observations	1728	1389		1728	1389	
R-squared	0.697	0.712		0.697	0.715	

5.5.2. Non-monotonicity

In this subsection, I exploit the theoretical non-monotonicity in the relation between market share and strategic behavior. Intuitively, a firm will use strategic mechanisms to curtail competition only if both of the following are jointly true: (1) the firm has enough market share that its actions materially affect the product market; and (2) the firm faces enough competitive pressure to care about curtailing it. Up until this point, the analysis has centered on the first condition. I now consider both conditions jointly.

To do so, I revisit the analyses in Table 3, but partition the sample into regions where the predicted marginal effect of market share is positive versus negative. In these analyses, I use raw market shares, rather than logged market shares, since the log transform differentially affects large versus small market shares.

Table 9 presents the results of these regressions. In Panel A (Panel B), I use a market share of 33.3% (50%) as the partitioning threshold. Across both panels, I document a similar pattern. In Cournot industries, the estimated coefficient on $Share \times PostCD\&A$ is positive (negative) for values of *Share* below (above) the threshold. In Bertrand industries, the estimated coefficients are all indistinguishable from zero. The differences between Cournot and Bertrand specifications are significant in three out of the four cases; it loses significance in the above 50% market share sample (perhaps due to the ~92.5% loss in sample size).

5.5.3. Product market outcomes

The preceding analyses are geared towards showing that firms respond, strategically, to a new pay package disclosure mandate. Theory predicts that large-share Cournot

firms will add revenue-based pay into their agents' pay packages—a notion that my empirical analyses corroborate. Theory also predicts that the disclosure of pay package details (and resulting changes in agents' incentives) will affect product market equilibria. Fershtman and Judd (1987) show that in concentrated Cournot product markets, firms' strategic use of revenue-based pay will result in more aggressive product market equilibria, characterized by increased production and lower margins/markups. I test this prediction using variations on the following regression specification:

$$\begin{aligned} <Outcome>_{i,t} = \beta_1 \log HHI_i \times PostCD\&A_t \times Cournot_i \\ &+ \beta_2 \log HHI_i \times PostCD\&A_t + \mu_i + \tau_t + \varepsilon_{i,t}, \end{aligned} \quad (11)$$

where *HHI* is the Herfindahl-Hirschman Index, calculated as a pre-CD&A average at the 4-digit SIC level. I examine three outcome variables: total production costs (*COGS*), gross profit to *COGS* (*Margin*), and EBITDA to sales (*Markup*). The results are tabulated in Table 10. For each outcome variable, I present two specifications. Odd-numbered specifications include the full sample (from 2002 to 2009), even-numbered specifications include only a narrow window around CD&A adoption, spanning from 2005 to 2007.

Across all six specifications, a consistent story emerges: in concentrated Cournot industries, total production expenditure increases, while margins/markups decrease. Collectively, these patterns are consistent with a post-CD&A shift towards revenue-based pay fostering more aggressive product market equilibria; firms expand production, resulting in lower market prices.

Table 9

Non-monotonicity.

This table presents evidence of the non-monotonic relation between market share and Cournot firms' shift towards revenue-based pay. Each specification is based on the regression:

$$\text{Revenue-Based Pay}_{i,t} = \beta \text{Share}_i \times \text{PostCD\&A}_t + \mu_i + \tau_t + \varepsilon_{i,t}.$$

Within each panel, I present the results of two sets of regressions, one set where *Share* is below a threshold value and one set where *Share* is above a threshold value. Panels differ only with respect to the threshold value chosen. In Panel A (Panel B), the sample is split at a market share of one-third (one-half). In odd-numbered (even-numbered) specifications, the sample is Cournot (Bertrand) industries. After each pair of regressions, I present the results of a statistical test of the difference across specifications. Below each coefficient estimate, I report a *t*-statistic calculated using robust standard errors clustered by firm. *, **, and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. All significance levels are based on two-tailed tests.

Panel A: Splitting at 33.3% market share						
Outcome = Revenue-Based Pay						
Variables	Less than one-third market share			More than one-third market share		
	(1)	(2)	(1) – (2)	(3)	(4)	(3) – (4)
<i>Share</i> × <i>PostCD&A</i>	0.984*** (3.141)	–0.248 (–0.553)	1.232** (2.259)	–0.290* (–1.992)	0.292 (1.266)	–0.582** (–2.132)
Competition Type	Cournot	Bertrand		Cournot	Bertrand	
Firm Fixed Effects	Yes	Yes		Yes	Yes	
Year Fixed Effects	Yes	Yes		Yes	Yes	
Observations	1520	1130		208	259	
R-squared	0.691	0.705		0.785	0.759	
Panel B: Splitting at 50% market share						
Outcome = Revenue-Based Pay						
Variables	Less than one-half market share			More than one-half market share		
	(1)	(2)	(1) – (2)	(3)	(4)	(3) – (4)
<i>Share</i> × <i>PostCD&A</i>	0.614*** (3.215)	–0.129 (–0.638)	0.743*** (2.675)	–0.591* (–1.949)	0.062 (0.116)	–0.653 (–1.073)
Competition Type	Cournot	Bertrand		Cournot	Bertrand	
Firm Fixed Effects	Yes	Yes		Yes	Yes	
Year Fixed Effects	Yes	Yes		Yes	Yes	
Observations	1609	1270		119	119	
R-squared	0.694	0.713		0.744	0.652	

Table 10

Product market outcomes.

This table presents evidence on the differential shifts in product market outcomes across Cournot and Bertrand industries, around the introduction of the CD&A. All specifications are based on the following regression:

$$< \text{Outcome} >_{i,t} = \beta_1 \log HHI_i \times \text{PostCD\&A}_t \times \text{Cournot}_i + \beta_2 \log HHI_i \times \text{PostCD\&A}_t + \mu_i + \tau_t + \varepsilon_{i,t},$$

where *HHI* is the Herfindahl-Hirschman Index, calculated as a pre-CD&A average at the 4-digit SIC level. I examine three outcome variables: total production costs (COGS) in Specifications 1 and 2; gross margins in Specifications 3 and 4; and markups (EBITDA to sales), in Specifications 5 and 6. For each outcome variable, I present two specifications which differ only with respect to sample window. In odd-numbered (even-numbered) specifications I use the full sample window from 2002 to 2009 (a restricted sample window from 2005 to 2007). Below each coefficient estimate, I report a *t*-statistic, calculated using standard errors clustered by firm. *, **, and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. All significance levels are based on two-tailed tests.

	Outcome = <i>ln</i> (COGS)		Outcome = <i>ln</i> (Margin)		Outcome = <i>ln</i> (Markup)	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>ln</i> (HHI) × <i>PostCD&A</i> × <i>Cournot</i>	0.154*** (2.903)	0.098*** (3.036)	–0.140*** (–3.471)	–0.154*** (–3.826)	–0.111** (–2.459)	–0.127*** (–3.038)
<i>ln</i> (HHI) × <i>PostCD&A</i>	–0.117*** (–2.600)	–0.066** (–2.290)	0.126*** (3.867)	0.105*** (3.242)	0.106*** (2.768)	0.080** (2.290)
Sample Window	Full	2005 – 2007	Full	2005 – 2007	Full	2005 – 2007
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year-Mode Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2654	1041	2627	1033	2548	1009
R-squared	0.964	0.994	0.948	0.983	0.821	0.925

5.6. Assessing the alternative: Pre-CD&A non-disclosure

My data on performance metrics come from firms' proxy disclosures. If these disclosures do not perfectly reveal the actual metrics used for executive performance evaluation, it is possible that the apparent changes in pay practices do not represent actual changes in pay practices, but merely changes in disclosed pay practices. In this case, my results would suggest that oligopolistic Cournot firms were already using revenue-based pay, but choosing to hide its use until the CD&A made such obfuscation more costly. In what follows, I refer to this possibility as the "non-disclosure interpretation."

I take several steps to address the possibility that pre-CD&A non-disclosure explains my results. First and foremost, in my sample construction, I drop any observations for which zero metrics are listed, as well as any observations for which Incentive Lab indicates that metric choices were "Vague," which represents about 8% of pre-CD&A firm-years. Thus, my sample includes only those observations for which at least one metric is clearly disclosed and none are vaguely disclosed. If pre-CD&A non-disclosure explains my results, it would require selective non-disclosure of revenue-based pay, especially among oligopolistic Cournot firms.

To further assess the plausibility of the non-disclosure interpretation, I perform two additional diagnostic tests: (1) I estimate what the pre-CD&A non-disclosure rate would have to have been to generate my findings; and (2) I examine the rate at which firms remove (as opposed to add) disclosed metrics after CD&A adoption. In both tests, I focus on December year-end firms, as these are the firms for which advanced knowledge of the impending disclosure rules is most questionable.

As an initial diagnostic test, I calculate the implied pre-CD&A non-disclosure rate under the assumption that all apparent additions of revenue-based pay are the result of firms finally disclosing the use of revenue-based pay that had already been present in the CEO's pay package. To do so, I count: (1) the number of firms disclosing the use of revenue-based pay in the last year of the pre-CD&A period ($\#Disclosed$); and (2) the number of firms not disclosing the use revenue-based pay in the last year of the pre-CD&A period, but disclosing the use of revenue-based pay in the first year of the post-CD&A period ($\#Undisclosed$). Under the non-disclosure interpretation, these represent firms whose usage of revenue-based pay was previously undisclosed. I then calculate the implied non-disclosure rate as the ratio $\frac{\#Undisclosed}{\#Disclosed + \#Undisclosed}$.

I find that, under the non-disclosure interpretation, pre-CD&A non-disclosure would have to have been a pervasive issue; the implied non-disclosure rate is roughly 31%. That is, under the non-disclosure interpretation, almost one-third of the firms using revenue-based pay were hiding its use, while clearly disclosing their reliance on other performance metrics. It is not obvious why firms would seek to selectively obscure their reliance on revenue-based pay. Moreover, the implied non-disclosure rate is particularly high for oligopolistic Cournot firms (i.e., those with market shares between 10% and 65%). Under the non-disclosure interpretation, the majority (53%) of these firms using

revenue-based pay were hiding its use. By comparison, similarly oligopolistic Bertrand firms and less oligopolistic Cournot firms have implied pre-CD&A non-disclosure rates of 22% and 24%, respectively. These patterns do not appear to support the non-disclosure interpretation. It seems unlikely that selective obfuscation of revenue-based pay would be so prevalent, and it is especially unlikely that oligopolistic Cournot firms—those that stand to benefit most from highlighting, rather than obfuscating, their use of revenue-based pay—would be the worst offenders.

As a second diagnostic test, I leverage the inherently asymmetric effect the non-disclosure interpretation implies; better disclosures under the CD&A could result in the apparent addition of new metrics, but not the apparent removal of old ones. In contrast, if the introduction of the CD&A causes firms to re-evaluate/adjust their compensation practices, this could result in both additions and removals. In line with this intuition, I examine the rates at which revenue-based pay gets added to and removed from executive compensation plans, over the years leading up to and including the first year of the post-CD&A period.

I find that, in their first year of CD&A compliance, December year-end Cournot firms appear to both add and remove revenue-based pay at substantially greater rates than in the prior years. Relative to the preceding three years (2003–2005), the apparent addition and removal rates for revenue metrics increase by 220% and 170%, respectively. While the net effect is an increase in the disclosed use of revenue-based pay, the rate at which firms appear to remove revenue-based pay from their CEOs' pay packages nearly triples in the first year of CD&A compliance. Consistent with the Fershtman and Judd strategy, more (less) oligopolistic Cournot firms drive the apparent increase in addition rates (removal rates). While an apparent uptick in metric additions can be explained by a mechanical effect of better disclosure, an apparent uptick in metric removals cannot. In sum, the patterns present in the data favor an interpretation whereby firms adjust their actual compensation practices around CD&A adoption. While I cannot definitively refute the non-disclosure interpretation, the data do not appear to support it.

5.7. Sensitivity analyses

To ensure that my results are robust to alternative design choices, I replicate my primary analyses with several modifications.

5.7.1. Alternative measures for the mode of competition

Modes of strategic conduct are notoriously difficult to evaluate, and no single measure is likely to capture them perfectly. Hence, I construct seven additional measures for the mode of competition (Flex., R&D, Quant., Pricing, Mines, Vote, and Factor) to better triangulate the underlying construct. I replicate the main analysis using all eight measures, and present the results in Table A.2, in the Appendix. Across all eight codings, I find a consistent pattern: large-share Cournot competitors are significantly more likely to begin using revenue-based pay in their CEOs' pay packages after the introduction of the CD&A. This pattern is absent among Bertrand competitors. The

difference across Cournot and Bertrand industries is economically large and statistically significant.

5.7.2. Alternative measures of product market influence

In my primary analyses, I use $\ln(\text{Share})$ to reflect a firm's product market influence. To ensure that my results are robust to different measurement approaches, I construct six alternative measures. The first two are also continuous transformations of Share : Share^γ , where $\gamma = 0.149521$ was numerically determined to be the absolute-skewness minimizing parameter; and $\sqrt{\text{Share}}$. The next alternative measure is Share , without any transformation. The final three alternative measures are indicator variables of product market influence: Leader , Top2 , and Top3 . These indicator variables reflect whether the firm has the highest, one of the two highest, and one of the three highest market shares in the 4-digit SIC.

I replicate Table 3 using all six alternative measures and present the results in Table IA1, in the Internet Appendix. Across all six alternative measures, I find that larger-share Cournot firms are significantly more likely than smaller-share Cournot firms to add revenue-based pay to their CEOs' pay packages after CD&A adoption. I do not find the same to be true among Bertrand firms for any measure. Moreover, the difference between Cournot and Bertrand firms is statistically significant for all but two measures: Share and Leader . Both insignificant differences are driven by extremely large-share outliers; in untabulated analyses, I find that both differences become statistically significant after dropping observations with >90% market shares (~1.5% of the sample).

5.7.3. Balanced panel analyses

To ensure that my results are not driven by changing sample composition, I replicate my analyses on a balanced panel of firms over a three-year period centered around CD&A adoption: 2005 to 2007. The balanced panel analyses yield the same inferences as the full sample tests, as shown in the Internet Appendix, Table IA2.

5.7.4. Logit specification

I use a linear probability model as my primary specification. This approach confers advantages in the form of easier interpretability and greater stability with dense fixed effect structures and many interactions (Neyman and Scott, 1948; Lancaster, 2000; Ai and Norton, 2003). In particular, linear probability models are robust to the "incidental parameters problem." However, given the well-documented issues associated with linear probability models (e.g., Maddala, 1986; Horrace and Oaxaca, 2006), I use logit analyses to verify that my results are not sensitive to the specification. My inferences are not affected by this alteration, as shown in the Internet Appendix, Table IA3.

5.7.5. Fiscal year-end exclusions

In the preceding analyses, I follow prior studies and focus on firms with September through December year ends (e.g., Ferri et al., 2018; Gipper, 2021). To ensure that this sample restriction does not drive my results, I replicate the main analysis using two alternative approaches: (1) retaining all firms, regardless of fiscal year-end; and (2) dropping

all firms with non-December year-ends. I document similar results, irrespective of this sample restriction, as shown in the Internet Appendix, Table IA4.

6. Conclusion

In this study, I provide initial evidence that some firms structure their CEOs' pay packages as strategic commitment devices. To do so, I exploit the introduction of a pay package disclosure mandate (the CD&A) as a source of plausibly exogenous variation in the benefits of strategic contracting, and examine how contractual terms in CEO pay plans change in response to the mandate.

I document that, immediately after the mandate, CEOs are more likely to have revenue-based pay added into their pay package if the benefits of aggressive commitment are greater. Moreover, I find evidence suggesting that this strategy affects product market outcomes. The shift towards revenue-based pay coincides with a shift towards more aggressive product market equilibria, characterized by greater levels of production and lower margins/markups. Collectively, these results provide some of the only broad sample archival evidence to support the theoretical literature on "strategic delegation" (e.g., Schelling, 1960; Fershtman and Judd, 1987; Katz, 1991). My findings do not refute existing studies on the determinants of executive pay. Instead, my results add an additional determinant to the list by highlighting a relatively unexplored role of executive compensation contracts: devices to facilitate strategic commitment.

Appendix A. Measuring the mode of competition

In this Appendix, I discuss the seven alternative measures used to estimate the mode of competition. Codings of the Fama and French 48 industries for each measure are presented in Table A.1. The main results using each measure are tabulated in Table A.2.

Flex. measure

The first supplemental measure is based on production flexibility (i.e., the importance of capacity constraints). Kreps and Scheinkman (1983) show that Bertrand competition is equivalent to Cournot competition if firms must commit to quantities through capacity investments. Dixon (1986) elaborates:

"It has long been recognized that flexibility of production lies at the heart of the distinction between Bertrand and Cournot models. The most natural application of the Cournot model would seem to be in the case where output is fixed in the short run."

Maggi (1996) formalizes this intuition with an analytical model in which the mode of competition is determined endogenously by a single structural parameter: the importance of capacity constraints. If capacity constraints are not important, firms will endogenously choose to engage in Bertrand competition. If capacity constraints are more binding, firms will endogenously choose to engage in

Table A1

Fama-French 48 industries classified as Cournot or Bertrand.

This table presents the codings of each of the Fama and French industries as Cournot vs. Bertrand. For each industry, I present eight codings. “Kedia” codes industries using the measure developed by [Kedia \(2006\)](#). “Flex.” codes industries as Cournot (Bertrand) if industry-level average PP&E scaled by average assets is above (below) the median. “R&D” codes industries as Cournot (Bertrand) if industry-level average R&D spending scaled by SG&A is below (above) median. “Quant.” codes industries as Cournot (Bertrand) if industry-level average 8-K references to “production quantity” are above (below) median. “Pricing” codes industries as Cournot if, in an industry-specific timeseries regression, profit margins and HHI are significantly positively associated, and all other industries as Bertrand. “Mines” codes mining industries as Cournot, and all other industries as Bertrand. “Vote” codes industries as Cournot (Bertrand) if four or more (three or fewer) of the above industry classifications code the industry as Cournot (Bertrand). “Factor” codes industries as Cournot (Bertrand) if the industry’s ‘factor value’ is above (below) median. Panel B presents the factor loadings for the principal factor of the six non-aggregate measures (Kedia, Flex, R&D, Quant, Pricing and Mines).

Panel A: Classification of industries as Cournot versus Bertrand for all eight measures									
Ind.	Description	Kedia	Flex.	R&D	Quant.	Pricing	Mines	Vote	Factor
1	Agriculture	Cournot	Cournot	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand
2	Food Products	Bertrand	Cournot	Cournot	Cournot	Cournot	Bertrand	Cournot	Cournot
3	Candy & Soda	Cournot	Cournot	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand
4	Beer & Liquor	Cournot	Cournot	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand
5	Tobacco Products	Bertrand	Bertrand	Cournot	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand
6	Recreation	Bertrand	Bertrand	Cournot	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand
7	Entertainment	Bertrand	Cournot	Cournot	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand
8	Printing and Publishing	Cournot	Cournot	Cournot	Cournot	Bertrand	Bertrand	Cournot	Cournot
9	Consumer Good	Bertrand	Cournot	Cournot	Cournot	Cournot	Bertrand	Cournot	Cournot
10	Apparel	Bertrand	Bertrand	Cournot	Cournot	Cournot	Bertrand	Bertrand	Cournot
11	Healthcare	Cournot	Cournot	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand
12	Medical Equip.	Bertrand	Bertrand	Cournot	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand
13	Pharmaceutical Products	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand
14	Chemicals	Bertrand	Cournot	Cournot	Cournot	Bertrand	Bertrand	Bertrand	Cournot
15	Rubber and Plastic Products	Cournot	Cournot	Cournot	Cournot	Bertrand	Bertrand	Cournot	Cournot
17	Construction Materials	Bertrand	Cournot	Cournot	Cournot	Cournot	Bertrand	Cournot	Cournot
18	Construction	Bertrand	Bertrand	Cournot	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand
19	Steel Works Etc	Bertrand	Cournot	Cournot	Cournot	Bertrand	Bertrand	Bertrand	Cournot
21	Machinery	Cournot	Bertrand	Cournot	Cournot	Cournot	Bertrand	Cournot	Cournot
22	Electrical Equipment	Bertrand	Cournot	Cournot	Cournot	Bertrand	Bertrand	Bertrand	Cournot
23	Automobiles and Trucks	Bertrand	Cournot	Bertrand	Cournot	Bertrand	Bertrand	Bertrand	Bertrand
24	Aircraft	Bertrand	Cournot	Bertrand	Cournot	Bertrand	Bertrand	Bertrand	Bertrand
25	Shipbuilding, Railroad Equip.	Cournot	Cournot	Cournot	Bertrand	Bertrand	Bertrand	Bertrand	Cournot
27	Precious Metals	Cournot	Cournot	Bertrand	Cournot	Bertrand	Cournot	Cournot	Cournot
28	Non-Met. and Indust. Mining	Cournot	Cournot	Cournot	Cournot	Bertrand	Cournot	Cournot	Cournot
29	Coal	Cournot	Cournot	Bertrand	Cournot	Bertrand	Cournot	Cournot	Cournot
30	Petroleum and Natural Gas	Cournot	Cournot	Cournot	Cournot	Cournot	Bertrand	Cournot	Cournot
31	Utilities	Cournot	Cournot	Bertrand	Cournot	Cournot	Bertrand	Cournot	Cournot
32	Communication	Cournot	Cournot	Cournot	Bertrand	Cournot	Bertrand	Cournot	Cournot
33	Personal Services	Bertrand	Cournot	Bertrand	Cournot	Bertrand	Bertrand	Bertrand	Bertrand
34	Business Services	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand
35	Computers	Cournot	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand
36	Electronic Equipment	Bertrand	Bertrand	Bertrand	Cournot	Bertrand	Bertrand	Bertrand	Bertrand
37	Measuring and Control Equip	Bertrand	Bertrand	Bertrand	Cournot	Bertrand	Bertrand	Bertrand	Bertrand
38	Business Supplies	Cournot	Cournot	Cournot	Cournot	Bertrand	Bertrand	Cournot	Cournot
39	Shipping Containers	Cournot	Cournot	Cournot	Bertrand	Bertrand	Bertrand	Bertrand	Cournot
40	Transportation	Cournot	Cournot	Bertrand	Cournot	Cournot	Bertrand	Cournot	Cournot
41	Wholesale	Cournot	Bertrand	Cournot	Bertrand	Cournot	Bertrand	Bertrand	Cournot
42	Retail	Cournot	Cournot	Cournot	Cournot	Cournot	Bertrand	Cournot	Cournot
43	Restaurants	Bertrand	Cournot	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand
44	Banking	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand
45	Insurance	Cournot	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand
46	Real Estate	Cournot	Bertrand	Bertrand	Cournot	Cournot	Bertrand	Bertrand	Cournot
47	Trading	Cournot	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand	Bertrand
48	Almost Nothing	Cournot	Cournot	Bertrand	Cournot	Bertrand	Bertrand	Bertrand	Cournot
Panel B: Loadings on Principal Factor									
Measure								Factor Loading	
Kedia								0.369	
Flex.								0.683	
R&D								0.569	
Quant								0.669	
Pricing								0.748	
Mines								0.080	
Eigenvalue								1.940	

Table A.2

Main result across all eight measures for the mode of competition.

This table presents replications of the main result (Table 3) using all eight measures for the mode of competition. In Panel A, I use the Kedia measure; in Panel B, I use the Flex. measure; in Panel C, I use the R&D measure; in Panel D, I use the Quant. measure; in Panel E, I use the Pricing measure; in Panel F, I use the Mines measure; in Panel G, I use the Vote measure; in Panel H, I use the Factor measure. Within each panel, specifications differ only with respect to the sample. Specification 1 presents results for Cournot industries; Specification 2 presents results for Bertrand industries. The third column presents a test of the differences in coefficients across Specifications 1 and 2. Below each coefficient estimate, I report a *t*-statistic calculated using robust standard errors clustered by firm. *, **, and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. All significance levels are based on two-tailed tests.

Panel A: Kedia measure				Panel B: Flex. measure			
Variables	Outcome = Revenue-Based Pay			Variables	Outcome = Revenue-Based Pay		
	(1)	(2)	(1) – (2)		(1)	(2)	(1) – (2)
$\ln(\text{Share}) \times \text{PostCD\&A}$	0.034*** (2.753)	–0.016 (–1.308)	0.050*** (2.881)	$\ln(\text{Share}) \times \text{PostCD\&A}$	0.047*** (3.349)	–0.019* (–1.668)	0.066*** (3.648)
Competition Type	Cournot	Bertrand		Competition Type	Cournot	Bertrand	
Firm Fixed Effects	Yes	Yes		Firm Fixed Effects	Yes	Yes	
Year Fixed Effects	Yes	Yes		Year Fixed Effects	Yes	Yes	
Observations	1728	1389		Observations	1500	1617	
R-squared	0.694	0.707		R-squared	0.690	0.706	
Panel C: R&D measure				Panel D: Quant. measure			
Variables	Outcome = Revenue-Based Pay			Variables	Outcome = Revenue-Based Pay		
	(1)	(2)	(1) – (2)		(1)	(2)	(1) – (2)
$\ln(\text{Share}) \times \text{PostCD\&A}$	0.041*** (2.784)	–0.011 (–0.868)	0.051*** (2.688)	$\ln(\text{Share}) \times \text{PostCD\&A}$	0.026** (2.020)	–0.006 (–0.518)	0.032* (1.835)
Competition Type	Cournot	Bertrand		Competition Type	Cournot	Bertrand	
Firm Fixed Effects	Yes	Yes		Firm Fixed Effects	Yes	Yes	
Year Fixed Effects	Yes	Yes		Year Fixed Effects	Yes	Yes	
Observations	1274	1843		Observations	1539	1578	
R-squared	0.705	0.704		R-squared	0.672	0.710	
Panel E: Pricing measure				Panel F: Mines measure			
Variables	Outcome = Revenue-Based Pay			Variables	Outcome = Revenue-Based Pay		
	(1)	(2)	(1) – (2)		(1)	(2)	(1) – (2)
$\ln(\text{Share}) \times \text{PostCD\&A}$	0.035** (2.452)	–0.004 (–0.364)	0.039** (2.189)	$\ln(\text{Share}) \times \text{PostCD\&A}$	0.191* (1.824)	0.004 (0.404)	0.188** (2.018)
Competition Type	Cournot	Bertrand		Competition Type	Cournot	Bertrand	
Firm Fixed Effects	Yes	Yes		Firm Fixed Effects	Yes	Yes	
Year Fixed Effects	Yes	Yes		Year Fixed Effects	Yes	Yes	
Observations	1062	2055		Observations	60	3057	
R-squared	0.708	0.698		R-squared	0.502	0.706	
Panel G: Vote measure				Panel H: Factor measure			
Variables	Outcome = Revenue-Based Pay			Variables	Outcome = Revenue-Based Pay		
	(1)	(2)	(1) – (2)		(1)	(2)	(1) – (2)
$\ln(\text{Share}) \times \text{PostCD\&A}$	0.057*** (3.789)	–0.013 (–1.236)	0.070*** (3.823)	$\ln(\text{Share}) \times \text{PostCD\&A}$	0.047*** (3.479)	–0.016 (–1.393)	0.063*** (3.540)
Competition Type	Cournot	Bertrand		Competition Type	Cournot	Bertrand	
Firm Fixed Effects	Yes	Yes		Firm Fixed Effects	Yes	Yes	
Year Fixed Effects	Yes	Yes		Year Fixed Effects	Yes	Yes	
Observations	1117	2000		Observations	1521	1596	
R-squared	0.692	0.707		R-squared	0.679	0.712	

Cournot competition. In this spirit, I capture Cournot versus Bertrand competition using a measure of production flexibility based on capacity investments, and refer to this measure as “Flex.”

I measure production flexibility based on industry-level capital investment. I calculate the industry-level average ratio of PP&E to total assets, and code industries as Cournot (Bertrand) if the ratio is above (below) the median. Be-

cause capital investments are potentially endogenous to changes in executive incentives, I rely solely on pre-2006 data to classify industries.

R&D measure

The second supplemental measure is based on firms' R&D investments. Ample theoretical studies show that R&D is more prevalent in Bertrand industries than Cournot industries (e.g., [Delbono and Denicolò, 1990](#); [Motta, 1993](#); [Brander and Spencer, 2015](#); [Lee and Choi, 2016](#)). Furthermore, if firms can endogenously choose the mode of competition, lower levels of differentiation pushes firms to choose Cournot competition [Singh and Vives \(1984\)](#). Accordingly, I construct a measure based on the intensity of differentiation efforts (as captured by R&D investments), and refer to it as "R&D."

I calculate the industry-level average ratio of R&D spending to SG&A, and code industries as Cournot (Bertrand) if they are below (above) the median. Because R&D investments are potentially endogenous to changes in executive incentives, I rely solely on pre-2006 data to classify industries.

Quant. measure

The third supplemental measure is based on firms' mentions of "production quantities" in their 8-K filings. The underlying notion is that firms are more likely to discuss capacity/production quantities if these are actively made strategic choices, rather than passively realized equilibrium outcomes based on their other choices (e.g., product prices, à la Bertrand competition).

I use textual analysis to calculate the industry-level average frequency of "production quantity" references in firms' 8-K filings, using the same phrase list as [Bloomfield and Tuijn \(2019\)](#). I diverge from [Bloomfield and Tuijn \(2019\)](#) in that I do not make a directional requirement about increasing/decreasing. I code industries as Cournot (Bertrand) if the rate of production quantity mentions is above-median (below-median), and refer to this measure as "Quant."

Pricing measure

Prior studies suggest that the relation between margins and industry concentration is positive and linear in Cournot industries, but flat in Bertrand markets. (e.g., [Flath, 2012](#); [Waldman and Jensen, 2016](#)). With this in mind, I construct my fourth supplemental measure based on the relation between margins and HHI. For each 4-digit SIC, I regress gross margins on HHI. I code the industry as Cournot if at least one 4-digit SIC displayed a significantly positive association ($t > 1.96$). I refer to this measure as "Pricing."

Mines measure

The fifth supplemental measure codes only mining industries as Cournot. Extractive industries are well approximated by Cournot models, in that firms decide how much

to extract, and then realize an equilibrium price in the spot market (e.g., [Salant, 1976](#); [Sadorsky, 1992](#); [Ludkovski and Sircar, 2012](#); [Ludkovski and Sircar, 2015](#); [Trüby, 2013](#); [Hecking and Panke, 2015](#)).

I code firms as Cournot if they belong to Fama and French industries 27, 28 or 29 (precious metal mining; mineral and non-precious metal mining; coal mining), and refer to this measure as "Mines." This measure trades off Type I and Type II errors more conservatively than the others; it likely correctly codes all mining industries as Cournot, but also incorrectly codes many non-mining industries as Bertrand. This biases away from finding any results.

Vote measure

The sixth supplemental measure is a simple "majority vote" aggregation of the other measures. Industries are coded as Cournot if four or more of the preceding six measures (including Kedia) code it as such. I refer to this measure as "Vote."

Factor measure

The seventh supplemental measure is a factor analysis aggregation of the other measures (excluding Vote). I use factor analysis to identify the latent factor(s) underlying the six measures, and find that there is only one such factor with an eigenvalue greater than one,²² and it carries a positive weight from all six measures. The factor weights are presented in [Table A.1](#), Panel B. Using these factor weights, I calculate the "factor value" for each Fama and French industry, as the dot product of the six factor weights and the six corresponding indicator variables for Cournot classification. To convert to a binary measure, I code industries as Cournot (Bertrand) if the factor value is above median (below median).

References

- [Aggarwal, R.K., Samwick, A.A., 1999.](#) Executive compensation, strategic competition, and relative performance evaluation: theory and evidence. *J. Financ.* 54 (6), 1999–2043.
- [Ai, C., Norton, E.C., 2003.](#) Interaction terms in logit and probit models. *Econ. Lett.* 80 (1), 123–129.
- [Antón, M., Ederer, F., Giné, M., Schmalz, M.C., 2020.](#) Common ownership, competition, and top management incentives. *Ross Sch. Bus. Pap.* (1328).
- [Baiman, S., Demski, J.S., 1980.](#) Economically optimal performance evaluation and control systems. *J. Account. Res.* 184–220.
- [Baldenius, T., Michaeli, B., 2017.](#) Investments and risk transfers. *Account. Rev.* 92 (6), 1–23.
- [Baldenius, T., Michaeli, B., 2019.](#) Integrated ownership and managerial incentives with endogenous project risk. *Rev. Account. Stud.* 24 (4), 1450–1485.
- [Bertomeu, J., Evans, J.H., Feng, M., Tseng, A., et al., 2021.](#) Tacit collusion and voluntary disclosure: theory and evidence from the us automotive industry. *Manag. Sci.* 67 (3), 1851–1875.
- [Bloomfield, M.J., Marvão, C., Spagnolo, G., 2021.](#) Relative performance evaluation, sabotage and collusion. Available at SSRN 3524522.
- [Bloomfield, M. J., Gipper, B., Kepler, J. D., Tsui, D., 2020.](#) Cost shielding in executive bonus plans. Available at SSRN 3525148.

²² The eigenvalue is 1.940, and the next highest has an eigenvalue of 0.326.

- Bloomfield, M. J., Tuijn, M., 2019. Do firms strategically announce capacity expansions in response to heightened entry threats? Available at SSRN 3195932.
- Bourveau, T., She, G., Žaldokas, A., 2020. Corporate disclosure as a tacit coordination mechanism: evidence from cartel enforcement regulations. *J. Account. Res.* 58 (2), 295–332.
- Brander, J.A., Lewis, T.R., 1986. Oligopoly and financial structure: the limited liability effect. *Am. Econ. Rev.* 76 (5), 956–970.
- Brander, J.A., Lewis, T.R., 1988. Bankruptcy costs and the theory of oligopoly. *Can. J. Econ.* 21 (2), 221–243.
- Brander, J.A., Spencer, B.J., 1983. Strategic commitment with R&D: the symmetric case. *Bell J. Econ.* 14 (1), 225–235.
- Brander, J.A., Spencer, B.J., 2015. Endogenous Horizontal Product Differentiation under Bertrand and Cournot Competition: Revisiting the Bertrand Paradox. Technical Report. National Bureau of Economic Research.
- Bresnahan, T.F., 1989. Empirical studies of industries with market power. In: *Handbook of Industrial Organization*, 2. Elsevier, pp. 1011–1057.
- Burks, J.J., Cuny, C., Gerakos, J., Granja, J., 2018. Competition and voluntary disclosure: evidence from deregulation in the banking industry. *Rev. Account. Stud.* 23 (4), 1471–1511.
- Coles, J.L., Daniel, N.D., Naveen, L., 2006. Managerial incentives and risk-taking. *J. Financ. Econ.* 79 (2), 431–468.
- Cournot, A.-A., 1838. Recherches sur les principes mathématiques de la théorie des richesses par Augustin Cournot. chez L. Hachette.
- Delbono, F., Denicolò, V., 1990. R & D investment in a symmetric and homogeneous oligopoly: Bertrand vs Cournot. *Int. J. Ind. Organiz.* 8 (2), 297–313.
- Dittmann, I., Yu, K.-C., Zhang, D., 2017. How important are risk-taking incentives in executive compensation? *Rev. Financ.* 21 (5), 1805–1846.
- Dixit, A., 1980. The role of investment in entry-deterrence. *Econ. J.* 90 (357), 95–106.
- Dixon, H., 1986. The Cournot and Bertrand outcomes as equilibria in a strategic metagame. *Econ. J.* 96, 59–70.
- Ferri, F., Zheng, R., Zou, Y., 2018. Uncertainty about managers' reporting objectives and investors' response to earnings reports: evidence from the 2006 executive compensation disclosures. *J. Account. Econ.* 66 (2–3), 339–365.
- Fershtman, C., 1985. Managerial incentives as a strategic variable in duopolistic environment. *Int. J. Ind. Org.* 3 (2), 245–253.
- Fershtman, C., Judd, K.L., 1987. Equilibrium incentives in oligopoly. *Am. Econ. Rev.* 927–940.
- Flath, D., 2012. Are there any Cournot industries in Japan? *Jpn. Econ.* 39 (2), 3–36.
- Gipper, B., 2021. The economic effects of expanded compensation disclosures. *J. Account. Econ.* 71 (1), 101–338.
- Glaeser, S., Landsman, W. R., 2019. Deterrent disclosure. Available at SSRN 3342691.
- Hart, O., Tirole, J., Carlton, D.W., Williamson, O.E., 1990. Vertical integration and market foreclosure. *Brook. Pap. Econ. Act. Microecon.* 1990, 205–286.
- Hecking, H., Panke, T., 2015. The global markets for coking coal and iron ore—complementary goods, integrated mining companies and strategic behavior. *Energy Econ.* 52, 26–38.
- Holmström, B., 1979. Moral hazard and observability. *Bell J. Econ.* 74–91.
- Holmström, B., Milgrom, P., 1987. Aggregation and linearity in the provision of intertemporal incentives. *Econometrica: Journal of the Econometric Society* 303–328.
- Holmström, B., Milgrom, P., 1991. Multitask principal-agent analyses: incentive contracts, asset ownership, and job design. *J. Law, Econ. Org.* 7, 24–52.
- Holmström, B., Milgrom, P., 1994. The firm as an incentive system. *Am. Econ. Rev.* 972–991.
- Horrace, W.C., Oaxaca, R.L., 2006. Results on the bias and inconsistency of ordinary least squares for the linear probability model. *Econ. Lett.* 90 (3), 321–327.
- Jensen, M.C., Meckling, W.H., 1979. Theory of the firm: Managerial behavior, agency costs and ownership structure. *J. Financ. Econ.* 3 (4), 305–360.
- Katz, M.L., 1991. Game-playing agents: unobservable contracts as precommitments. *Rand. J. Econ.* 307–328.
- Kedia, S., 2006. Estimating product market competition: methodology and application. *J. Bank. Financ.* 30 (3), 875–894.
- Kepler, J.D., 2021. Private communication among competitors and public disclosure. *J. Account. Econ.* 101–387.
- Knopf, J.D., Nam, J., Thornton, J.H., 2002. The volatility and price sensitivities of managerial stock option portfolios and corporate hedging. *J. Financ.* 57 (2), 801–813.
- Koçkesen, L., 2007. Unobservable contracts as precommitments. *Econ. Theory* 31 (3), 539–552.
- Koçkesen, L., Ok, E.A., 2004. Strategic delegation by unobservable incentive contracts. *Rev. Econ. Stud.* 71 (2), 397–424.
- Kreps, D.M., Scheinkman, J.A., 1983. Quantity precommitment and Bertrand competition yield Cournot outcomes. *Bell J. Econ.* 326–337.
- Lambert, R.A., Larcker, D.F., 1987. An analysis of the use of accounting and market measures of performance in executive compensation contracts. *J. Account. Res.* 85–125.
- Lancaster, T., 2000. The incidental parameter problem since 1948. *J. Econ.* 95 (2), 391–413.
- Lee, D., Choi, K., 2016. Bertrand vs. Cournot competition with upstream firm investment. *Bull. Econ. Res.* 68 (S1), 56–65.
- Low, A., 2009. Managerial risk-taking behavior and equity-based compensation. *J. Financ. Econ.* 92 (3), 470–490.
- Ludkovski, M., Sircar, R., 2012. Exploration and exhaustibility in dynamic Cournot games. *Eur. J. Appl. Math.* 23 (3), 343–372.
- Ludkovski, M., Sircar, R., 2015. Game theoretic models for energy production. In: *Commodities, Energy and Environmental Finance*. Springer, pp. 317–333.
- Maddala, G.S., 1986. *Limited-Dependent and Qualitative Variables in Econometrics*. Cambridge University Press.
- Maggi, G., 1996. Strategic trade policies with endogenous mode of competition. *Am. Econ. Rev.* 237–258.
- McGuire, T.W., Staelin, R., 1983. An industry equilibrium analysis of downstream vertical integration. *Mark. Sci.* 2 (2), 161–191.
- Moreno, D., Ubeda, L., 2006. Capacity precommitment and price competition yield the Cournot outcome. *Games Econ. Behav.* 56 (2), 323–332.
- Morse, A., Nanda, V., Seru, A., 2011. Are incentive contracts rigged by powerful CEOs? *J. Financ.* 66 (5), 1779–1821.
- Motta, M., 1993. Endogenous quality choice: price vs. quantity competition. *J. Ind. Econ.* 113–131.
- Nevo, A., 1998. Identification of the oligopoly solution concept in a differentiated-products industry. *Econ. Lett.* 59 (3), 391–395.
- Neyman, J., Scott, E.L., 1948. Consistent estimates based on partially consistent observations. *Econometrika: J. Econom. Soc.* 1–32.
- Raith, M., 2003. Competition, risk and managerial incentives. *Am. Econ. Rev.* 93, 1425–1436.
- Sadorsky, P.A., 1992. Industry size and destructive competition in Cournot oligopoly models of exhaustible resource exploration and extraction. *Res. Energy* 14 (3), 249–257.
- Salant, S.W., 1976. Exhaustible resources and industrial structure: a Nash-Cournot approach to the world oil market. *J. Polit. Econ.* 84 (5), 1079–1093.
- Securities and Exchange Commission, 2016. About the SEC. <https://www.sec.gov/about.shtml>.
- Schelling, T.C., 1960. *The Strategy of Conflict*. Cambridge, Mass.
- Schwartz, M., Thompson, E.A., 1986. Divisionalization and entry deterrence. *Q. J. Econ.* 101 (2), 307–321.
- Securities and Exchange Commission, 2006a. Executive compensation and related party disclosure: Release No. 33–8655. <https://www.sec.gov/rules/proposed/33-8655.pdf>.
- Securities and Exchange Commission, 2006b. Executive compensation and related person disclosure: Release No. 33–8732a. <https://www.sec.gov/rules/final/2006/33-8732a.pdf>.
- Shue, K., Townsend, R.R., 2017. How do quasi-random option grants affect CEO risk-taking? *J. Financ.* 72 (6), 2551–2588.
- Singh, N., Vives, X., 1984. Price and quantity competition in a differentiated duopoly. *Rand. J. Econ.* 546–554.
- Skivas, S.D., 1987. The strategic choice of managerial incentives. *Rand. J. Econ.* 452–458.
- Spence, M., 1973. Job market signaling. *Q. J. Econ.* 87 (3), 355–374.
- Trüb, J., 2013. Strategic behaviour in international metallurgical coal markets. *Energy Econ.* 36, 147–157.
- Veendorp, E.C., 1991. Executive compensation and related party disclosure: release no. 33–8655. *Q. J. Econ.* 106 (1), 297–307.
- Vickers, J., 1985. Delegation and the theory of the firm. *Econ. J.* 95, 138–147.
- Waldman, D., Jensen, E., 2016. *Industrial Organization: Theory and Practice*. Routledge.